On Variance Reduction in Learning Mean Flows

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Abstract

Mean Flows learn promising distillation-free, one-step generation through optimizing the identity between average and instantaneous velocity field. It is, however, notorious for its non-decreasing, high-variance loss. We investigate the phenomenon and associates these manifestations to a fundamental issue in learning two-parameter average velocity field, that is, the existence of conditional-marginal velocity difference. Based on this discovery, we derive a concrete yet simple method, *Variance-Aware MeanFlow* (VaMF), to reduce the gradient variances and promote both efficiency and performance of the model. Empirically, VaMF improves the sliced Wasserstein-1 over vanilla MeanFlow by -23%, -13%, -23%, and -54% on `checkerboard`, `eight_gaussians`, `two_moons`, and `swiss_roll` respectively, dominates from $d = 4$ up to $d = 64$ on the dense-Gaussian-mixture scaling sweep, and transfers to large latent image generation models.

1 Introduction

Deep generative models that leverage deep neural networks to learn and generate samples from unknown data distributions have achieved profound success. One of the predominant paradigms adopts flow-based generative models [1, 2, 3, 4, 5, 6, 7] that learn continuous-time transformations between a simple prior and a target data distribution. The existing conditional flow matching [5, 6] and stochastic interpolant [7, 8] enable simulation-free training by regressing a velocity field, achieving sample quality competitive with diffusion models [9, 10, 11, 12, 13, 14, 15] and variational autoencoders [16]. However, flow-based models share the same limitation as diffusion models as they require iterative integration at inference time, which can suffer from extensive computational costs and numerical instability.

The issue motivates consecutive studies on reducing the number of function evaluations to one or few steps [17]. Existing approaches include progressive distillation [18], distribution matching [19, 20], consistency models [21, 22], and flow map matching [23], but most require a pre-trained teacher model. Recently, Mean Flows [24] offers an appealing *distillation-free* alternative by learning a two-parameter average velocity field $\boldsymbol{u}(\boldsymbol{x}, r, t)$ through a total-derivative identity between the average and marginal velocity field. In principle, this identity provides exact self-supervision. The compound prediction $\boldsymbol{u} + (t - r)\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{u}$ should match the instantaneous velocity at convergence, requiring no external teacher. Nevertheless, empirical training of Mean Flows is plagued by non-decreasing loss and prohibitively high gradient variance [25, 26].

In this work, we conduct a theoretical analysis that attributes the issue to replacing the marginal velocity fields by conditional fields in the total derivatives and the regression target, and establish the **curvature-variance principle**. Specifically, we prove that: **(i)** the per-step gradient variance is amplified by a Jacobi factor $\mathrm{Tr}(\boldsymbol{J}\Sigma_{\boldsymbol{v}'}\boldsymbol{J}^\top)$ where $\boldsymbol{J} = (t - r)\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}} - \boldsymbol{I}_d$, and the stop-gradient operator creates a semi-gradient gap that blinds the optimizer to this amplification; **(ii)** there exists a connection between the Jacobi and the curvature gap $\Delta := \boldsymbol{u} - \boldsymbol{v}$, which satisfies $\Delta \approx -\frac{(t-r)}{2}\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}$, so it is the local flow acceleration that causes the average and instantaneous velocities to diverge; and **(iii)** the acceleration-to-noise ratio $\mathrm{ANR} = (t - r)\|\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}\|/\sqrt{\mathrm{Tr}(\Sigma_{\boldsymbol{v}'})}$ governs the optimal sampling distribution over intervals, connecting scheduling heuristics in prior work to a principled criterion.

Building on this analysis, we propose **Variance-Aware MeanFlow (VaMF)**, a backbone-preserving training framework that addresses each diagnostic in turn. First, an *EMA velocity anchor* replaces the stochastic JVP tangent with a deterministic estimate, provably eliminating the Jacobi-amplified gradient noise. Then, a *flow-matching anchor* supervises the boundary condition at $r \to t$, removing the transient bias that an inexact tangent would otherwise introduce. Finally, a closed-form *trace weight* $w \propto 1/(1+\sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)/d)$ downweights each sample by the local Jacobian-deviation norm. VaMF preserves the original Mean Flow MSE objective and adds only one Jacobian-vector product per step. Our contributions are as follows:

C1 We establish the curvature-variance principle, which attributes the non-decreasing, high-variance loss of Mean Flows to a single mechanism—the Jacobi-amplified per-step gradient variance and its connection to the local flow acceleration—and provide the first formal quantification of the semi-gradient gap induced by the stop-gradient operator.

C2 We derive VaMF, a backbone-preserving training framework that combines an EMA velocity anchor, a flow-matching anchor, and a Hutchinson-estimated trace weight to eliminate variance amplification and the associated tangent bias.

C3 Empirically, VaMF improves Sliced Wasserstein-1 (SW_1) over vanilla MeanFlow by -23%, -13%, -23%, and -54% on the four 2D datasets, scales favorably from $d{=}2$ up to $d{=}64$ on the dense-Gaussian-mixture sweep, and transfers to large training of latent-space image generation models on the ImageNet datasets.

2 Preliminaries

Flow-Matching. Let $\mathcal{X} \subset \mathbb{R}^d$ be the space of random variables $\mathbf{x} \in \mathbb{R}^d$ and $\boldsymbol{x}$ a data sample as the realized state. The objective of flow-matching generative models is to learn a time-dependent diffeomorphic map $\psi(\boldsymbol{x}, t) : \mathbb{R}^d \times [0,1] \to \mathbb{R}^d$ from a simple distribution $\mathbf{x}_1 \sim p(\mathbf{x}, 1)$ to the target data distribution $\mathbf{x}_0 \sim p(\mathbf{x}, 0) = p_{\text{data}}$. Specifically, this is achieved by learning a *time-dependent velocity field* $\boldsymbol{v}(\boldsymbol{x}, t) : \mathbb{R}^d \times [0,1] \to \mathbb{R}^d$:

$$\boldsymbol{v}(\boldsymbol{x}, t) \triangleq \frac{\mathrm{d}}{\mathrm{d}t}\psi(\boldsymbol{x}, t), \quad \psi(\boldsymbol{x}, 0) = \boldsymbol{x}_0. \tag{1}$$

The map $\psi(\boldsymbol{x}, t)$ induces a *time-dependent probability density path* $p(\boldsymbol{x}, t) : \mathbb{R}^d \times [0,1] \to \mathbb{R}_{>0}$, with $\int_{\mathcal{X}} p(\boldsymbol{x}, t)\mathrm{d}\boldsymbol{x}$, $\forall t \in [0,1]$. However, the exact *marginal velocity field* $\boldsymbol{v}(\boldsymbol{x}, t)$ is generally *inaccessible*, especially in a high-dimensional sample space. The optimal-transport conditional flow-matching (OT-CFM) [5, 6] alternatively learns a conditional velocity field $\boldsymbol{v}(\boldsymbol{x}, t \mid \boldsymbol{x}_0; \boldsymbol{\theta}) \triangleq \boldsymbol{v}\left((1-t)\boldsymbol{x}_0 + t\boldsymbol{x}_1; \boldsymbol{\theta}\right)$. The surrogate loss substitution is justified by the following lemma:

Lemma 1. *Given vector fields $\boldsymbol{v}(\boldsymbol{x}, t \mid \boldsymbol{x}_0)$ that generate probability paths $p(\boldsymbol{x}, t \mid \boldsymbol{x}_0)$, for any data distribution $\mathbf{x}_0 \sim p(\mathbf{x}_0)$, the marginal velocity field is equal to the expected conditional velocity field*

$$\boldsymbol{v}(\boldsymbol{x}, t) = \mathbb{E}_{\boldsymbol{x}_0 \sim p(\boldsymbol{x}_0, t \mid \boldsymbol{x})}[\boldsymbol{v}(\boldsymbol{x}, t \mid \boldsymbol{x}_0)] \tag{2}$$

See full proof in appendix B.

One-step Mean Flows. Iterative integration of $\boldsymbol{v}(\boldsymbol{x}, t)$ is expensive and can be numerically unstable. To bypass it, Mean Flows [24] learn a two-parameter average velocity field $\boldsymbol{u}(\mathbf{x}, r, t)$ via the identity

$$(t-r)\,\boldsymbol{u}(\boldsymbol{x}, r, t) = \int_r^t \boldsymbol{v}(\boldsymbol{x}, \tau)\,\mathrm{d}\tau. \tag{3}$$

Differentiating with respect to the terminal time t yields

$$\boldsymbol{u}(\boldsymbol{x}, r, t) + (t-r)\,\tfrac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{u}(\boldsymbol{x}, r, t) = \boldsymbol{v}(\boldsymbol{x}, t), \qquad \tfrac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{u} = \partial_{\boldsymbol{x}}\boldsymbol{u}\cdot\boldsymbol{v}(\boldsymbol{x}, t) + \partial_t\boldsymbol{u}, \tag{4}$$

where the total derivative is a Jacobian-vector product (JVP), which we denote $\mathtt{JVP}(\boldsymbol{u}, (\boldsymbol{x}, r, t), (\boldsymbol{v}(\boldsymbol{x}, t), 0, 1))$. [24] replace both occurrences of the marginal field with the conditional field $\boldsymbol{v}_{\text{cond}} = \boldsymbol{x}_1 - \boldsymbol{x}_0$, giving the MeanFlow loss

$$\mathcal{L}_{\text{MF}}(\boldsymbol{\theta}) = \mathbb{E}_{r,t,\boldsymbol{x}_0,\boldsymbol{x}_1}\left[\|\boldsymbol{u}_{\boldsymbol{\theta}} + (t-r)\,\mathtt{sg}[\mathtt{JVP}(\boldsymbol{u}_{\boldsymbol{\theta}}, (\boldsymbol{z}, r, t), (\boldsymbol{v}_{\text{cond}}, 0, 1))] - \boldsymbol{v}_{\text{cond}}\|_2^2\right], \tag{5}$$

with $r, t \sim \mathcal{U}(0,1)$, $r \le t$, $\boldsymbol{x}_0 \sim p_{\text{data}}$, $\boldsymbol{x}_1 \sim \mathcal{N}(0, \boldsymbol{I}_d)$, $\boldsymbol{z} = (1-t)\boldsymbol{x}_0 + t\boldsymbol{x}_1$, and the stop-gradient $\mathtt{sg}[\cdot]$ avoiding double backpropagation through the JVP.

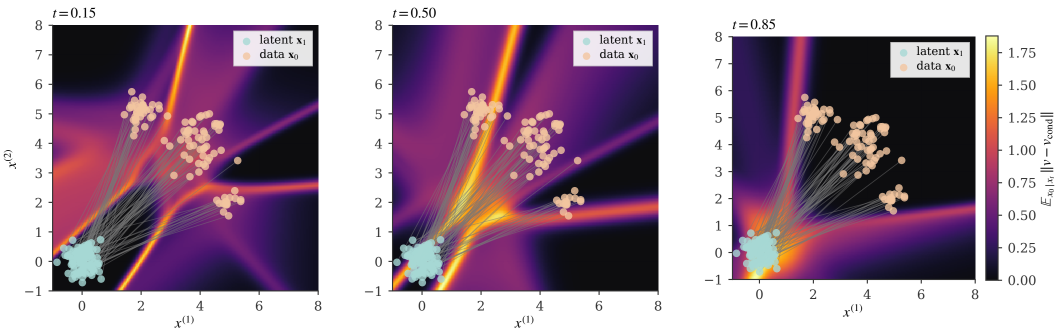

Figure 1: Conditional-marginal velocity gap on a 3-mode mixture. The spatial heatmap of $\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}\|\boldsymbol{v}(\boldsymbol{x}_t,t) - \boldsymbol{v}(\boldsymbol{x}_t,t \mid \boldsymbol{x}_0)\|$ at three timesteps confirms the gap is heterogeneous and concentrates in mode-mixing regions where multiple conditional paths overlap.

3 Variance-aware Mean Flows

The MeanFlow loss in equation 5 is empirically known to be *non-decreasing* and to suffer from *high-variance gradients* [25, 26]. In this section, we aim to answer the two complementary questions:

> ***RQ1* (Diagnosis):** What is the root cause non-decreasing, high-variance MeanFlow loss?

> ***RQ2* (Prescription):** Given the identified cause, what is the simplest modification that drives the gradient variance toward its irreducible level?

In the following, section 3.1 answers *RQ1* at the level of the gradients: a Reynolds decomposition exposes both the bias induced by stop-gradient training and the per-step gradient variance governed by the Jacobi factor $\boldsymbol{J} = (t-r)\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}} - \boldsymbol{I}_d$, and connects the bias to the *curvature gap* $\Delta = \boldsymbol{u} - \boldsymbol{v}$. Section 3.2 answers *RQ2* by proposing our training framework, *Variance-Aware MeanFlow (VaMF)*, comprising an EMA velocity anchor, a flow-matching anchor, and a closed-form trace weight, all applied on top of the original MSE objective without any architectural change. Full proofs are deferred to Appendix C.

3.1 Bias and Variances in Learning Mean Flows

We begin with investigating the consequences of replacing the inaccessible marginal velocity $\boldsymbol{v}(\boldsymbol{x},t)$ by the conditional field $\boldsymbol{v}_{\text{cond}} = \boldsymbol{x}_1 - \boldsymbol{x}_0$ in both the JVP tangent and the regression target. We first apply the Reynolds decomposition of the field and let $\boldsymbol{v}(\boldsymbol{x},t \mid \boldsymbol{x}_0) = \boldsymbol{v}(\boldsymbol{x},t) + \boldsymbol{v}'$, where $\boldsymbol{v}'$ is a zero-mean fluctuation from the marginal velocity induced by optimal transport between latent $\boldsymbol{x}_1$ and data $\boldsymbol{x}_0$. The regression loss equation 5 then expands to

$$\begin{aligned}\mathcal{L}_{\text{MF}}(\boldsymbol{\theta}) &= \mathbb{E}_{r,t,\boldsymbol{x}_0,\boldsymbol{x}_1}\left[\|\boldsymbol{u}_{\boldsymbol{\theta}} + (t-r)\mathtt{sg}\left[\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}}\cdot\boldsymbol{v}_{\text{cond}} + \partial_t\boldsymbol{u}_{\boldsymbol{\theta}}\right] - \boldsymbol{v}_{\text{cond}}\|_2^2\right] \\ &= \mathbb{E}_{r,t,\boldsymbol{x}_t,\boldsymbol{x}_1}\left[\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}\left[\left\|\underbrace{\boldsymbol{u}_{\boldsymbol{\theta}} + (t-r)\mathtt{sg}\left[\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}}\cdot\boldsymbol{v}(\boldsymbol{x}_t,t) + \partial_t\boldsymbol{u}_{\boldsymbol{\theta}}\right] - \boldsymbol{v}(\boldsymbol{x}_t,t)}_{\text{mean-field residual } \boldsymbol{r}^{\text{sg}}_{\boldsymbol{\theta}}(\boldsymbol{x}_t,t)} + \mathtt{sg}[\boldsymbol{J}]\boldsymbol{v}'\right\|_2^2\right]\right].\end{aligned} \tag{6}$$

Herein, we abuse the notations and let $\boldsymbol{J} \triangleq (t-r)\partial_{\boldsymbol{z}}\boldsymbol{u}_\theta - \boldsymbol{I}_d \in \mathbb{R}^{d\times d}$ denote a Jacobi factor of $\boldsymbol{u}_{\boldsymbol{\theta}}$ given sample $\boldsymbol{x}_t = \boldsymbol{x}_1 - \boldsymbol{x}_0$. This decomposition reveals that the original loss for learning Mean Flows is the sum of a deterministic mean-field residual $\boldsymbol{r}^{\text{sg}}_{\boldsymbol{\theta}}(\boldsymbol{x}_t,t)$ and a sample-based Jacobi-velocity-product $\text{sg}[\boldsymbol{J}]\boldsymbol{v}'$, where the former is the actual loss we aim to minimize given the identity in equation 4. If we further expand the inner expectation, we have $\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}[\boldsymbol{v}'] = 0$ based on equation 2 and the cross term vanishes, resulting in

$$\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}[\ell_{\text{MF}}(\boldsymbol{\theta})] = \mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}\left[\|\boldsymbol{r}(\boldsymbol{x}_t,t) + \mathtt{sg}[\boldsymbol{J}\boldsymbol{v}']\|_2^2\right] = \|\boldsymbol{r}_{\boldsymbol{\theta}}(\boldsymbol{x}_t,t)\|_2^2 + \mathtt{sg}[\text{Tr}(\boldsymbol{J}\Sigma_{\boldsymbol{v}'}\boldsymbol{J}^\top)], \tag{7}$$

where $\Sigma_{\boldsymbol{v}'} = \mathrm{Cov}_{\boldsymbol{x}_0|\boldsymbol{x}_t}[\boldsymbol{v}']$ is the conditional-velocity covariance and $\ell_{\mathrm{MF}}(\boldsymbol{\theta})$ is the per-sample loss. Replacing marginal velocity fields with conditional ones thus births an *irreducible per-sample variance* $\mathrm{Tr}(\Sigma_{\boldsymbol{v}'})$, which $\boldsymbol{J}$ amplifies into a per-step gradient variance growing quadratically in the spatial Jacobian.

Theorem 2 (Jacobian Variance Amplification)**.** *The Jacobi factor amplifies the gradient variance of the original MeanFlow loss:*

$$\boxed{\mathrm{Var}[\nabla_{\boldsymbol{\theta}}\ell_{MF}(\boldsymbol{\theta}) \mid \boldsymbol{x}_t] \;\propto\; \mathrm{Tr}\big(\boldsymbol{J}\,\Sigma_{\boldsymbol{v}'}\,\boldsymbol{J}^\top\big)\,.}$$

With large intervals $(t-r)$ or a large Jacobi norm, the per-sample gradient is dominated by the noise term $\boldsymbol{J}\boldsymbol{v}'$, downweighting the mean-field signal $\boldsymbol{r}_{\boldsymbol{\theta}}^{\mathrm{sg}}$. The proof is in Appendix C.

To help our readers better understand the existence of such a gap, we visualize the magnitude of this fluctuation $\|\boldsymbol{v}'\|$ on an example 3-Gaussian mixture in fig. 1. The gap is *non-zero*, *increases* when approaching the latent, and concentrates spatially in a *mode-mixing region near the latent* where multiple conditional paths overlap. Moreover, the stop-gradient operator creates a *semi-gradient gap*.

Theorem 3 (Semi-Gradient Gap)**.** *The gradients of the MeanFlow loss with and without the stop-gradient operator differ by*

$$\boxed{\underbrace{2(t-r)\,\mathbb{E}_{r,t,\boldsymbol{x}_0,\boldsymbol{x}_1}[\nabla_{\boldsymbol{\theta}}(\partial_{\boldsymbol{x}_t}\boldsymbol{u}_{\boldsymbol{\theta}}\cdot\boldsymbol{v}(\boldsymbol{x}_t,t)+\partial_t\boldsymbol{u}_{\boldsymbol{\theta}})\cdot\boldsymbol{r}_{\boldsymbol{\theta}}]}_{\textit{mean-field gradient difference}} + \underbrace{\mathbb{E}_{r,t,\boldsymbol{x}_0,\boldsymbol{x}_1}\big[\nabla_{\boldsymbol{\theta}}\,\mathrm{Tr}\big(\boldsymbol{J}\,\Sigma_{\boldsymbol{v}'}\,\boldsymbol{J}^\top\big)\big]}_{\textit{variance-driven gradient difference}}\,.}$$

At convergence the mean-field difference vanishes (since $\boldsymbol{r}_{\boldsymbol{\theta}}\to\boldsymbol{0}$) and the variance-driven term dominates: without stop-gradient it would force $\partial_{\boldsymbol{x}}\boldsymbol{u}_{\boldsymbol{\theta}} \to \frac{1}{t-r}\boldsymbol{I}_d$ to suppress $\boldsymbol{J}$, but with stop-gradient the optimizer is blind to $\mathrm{Tr}(\boldsymbol{J}\Sigma_{\boldsymbol{v}'}\boldsymbol{J}^\top)$, which can grow faster than $\|\boldsymbol{r}_{\boldsymbol{\theta}}\|_2^2$ shrinks, producing the non-decreasing loss reported empirically. The proof is in Appendix C.

To understand the root cause for both failure modes in theorem 2 and theorem 3, we define the *curvature gap* as the difference between the average and instantaneous velocity fields, by rearranging the identity in equation 4: $\Delta(\boldsymbol{x}_t,r,t) := \boldsymbol{u}(\boldsymbol{x}_t,r,t) - \boldsymbol{v}(\boldsymbol{x}_t,t) = -(t-r)\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{u}$. The mean-field residual in equation 6 can therefore be rewritten as $\boldsymbol{r}_{\boldsymbol{\theta}}^{\mathrm{sg}} = \Delta_{\boldsymbol{\theta}} + (t-r)\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{u}_{\boldsymbol{\theta}}$, where we simplify our notation and denote $\Delta_{\boldsymbol{\theta}} := \boldsymbol{u}_{\boldsymbol{\theta}} - \boldsymbol{v}$. At convergence the total derivative should exactly cancel the curvature gap, so the training signal $\boldsymbol{r}_{\boldsymbol{\theta}}^{\mathbf{sg}}$ is a small difference of two quantities whose magnitudes scale with $\|\Delta\|$. However, when $\|\Delta\|$ is large, this cancellation becomes delicate. Specifically, the gradient signal from $\boldsymbol{r}_{\boldsymbol{\theta}}$ is weak relative to the noise $\boldsymbol{J}\boldsymbol{v}'$, while the stop-gradient operator prevents the optimizer from controlling the resulting variance growth. This raises a natural question: how large is Δ in practice, and what governs its magnitude?

3.2 Variance-Aware MeanFlow

3.2.1 Target Network

In equation 6, the $\boldsymbol{J}\boldsymbol{v}'$ amplification arises because $\boldsymbol{v}'$ enters the JVP tangent, where it multiplies the spatial Jacobian $(t-r)\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}}$. We notice that, if the tangent were replaced by any *deterministic estimate* $\hat{\boldsymbol{v}}$, the fluctuation $\boldsymbol{v}'$ would only appear in the regression target, entering the residual linearly rather than through $\boldsymbol{J}$. In this case, the per-sample loss reduces to $\|\tilde{\boldsymbol{r}}_{\boldsymbol{\theta}}\|_2^2 + \mathrm{Tr}(\Sigma_{\boldsymbol{v}'})$, where $\tilde{\boldsymbol{r}}_{\boldsymbol{\theta}}$ is a deterministic residual. As a result, the variance drops to the irreducible level $\mathrm{Tr}(\Sigma_{\boldsymbol{v}'})$ without any Jacobian amplification. We higlight this as a theoretical foundation missing from the proposed method in a recent study [26]. The remaining question is whether we can achieve this without learning a separate velocity model.

We propose to replace the stochastic JVP tangent $\boldsymbol{v}_{\mathrm{cond}}$ with a deterministic estimate from a target network. Specifically, we use a Polyak averaging [27] (*a.k.a. exponential moving average (EMA)* [28]) of the model parameters as the target network, and the new loss writes

$$\mathcal{L}_{\mathrm{MF}}^{\mathrm{EMA}}(\boldsymbol{\theta}) = \mathbb{E}_{r,t,\boldsymbol{x}_0,\boldsymbol{x}_1}\left[\left\|\boldsymbol{u}_{\boldsymbol{\theta}} + (t-r)\mathtt{sg}\left[\mathtt{JVP}(\boldsymbol{u}_{\boldsymbol{\theta}},(\boldsymbol{z},r,t),(\boldsymbol{u}_{\bar{\boldsymbol{\theta}},t},0,1))\right] - \boldsymbol{v}_{\mathrm{cond}}\right\|_2^2\right], \tag{8}$$

where $\boldsymbol{u}_{\bar{\boldsymbol{\theta}},t} = \boldsymbol{u}(\boldsymbol{x}_t,t,t)$ and $\bar{\boldsymbol{\theta}}$ is updated by $\bar{\boldsymbol{\theta}} \leftarrow \mu\bar{\boldsymbol{\theta}} + (1-\mu)\boldsymbol{\theta}$ with $\mu \in [0,1)$.

3.2.2 Flow-matching Anchor

The EMA tangent helps resolve the Jacobian variance amplification but introduces a transient bias when $\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t, t, t) \neq \boldsymbol{v}(\boldsymbol{x}_t, t)$. To correct this, we add a *flow matching anchor* that provides direct velocity supervision at the boundary $r \to t$:

$$\mathcal{L}_{\text{FM}}(\boldsymbol{\theta}) = \mathbb{E}_{t,\boldsymbol{x}_0,\boldsymbol{x}_1}\left[\|\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, t{-}\delta, t) - \boldsymbol{v}_{\text{cond}}\|_2^2\right], \tag{9}$$

where $\delta \sim \mathcal{U}[\delta_{\min}, \delta_{\max}]$ with $\delta_{\max} \ll 1$. For small δ, the average velocity over $[t{-}\delta, t]$ approximates the instantaneous velocity. The full objective in VaMF is given by

$$\mathcal{L}_{\text{VaMF}}(\boldsymbol{\theta}) = \mathcal{L}_{\text{MF}}^{\text{EMA}}(\boldsymbol{\theta}) + \lambda\,\mathcal{L}_{\text{FM}}(\boldsymbol{\theta}), \tag{10}$$

where $\lambda > 0$ balances the two terms. We characterize the gradient and the residual structure under this objective with two propositions; both proofs are in Appendix C.

Proposition 4 (Gradient of VaMF Loss)**.** *The gradient of $\mathcal{L}_{MF}^{EMA}$ takes the form*

$$\nabla_{\boldsymbol{\theta}}\mathcal{L}_{MF}^{EMA} \;=\; 2\,\mathbb{E}\big[\nabla_{\boldsymbol{\theta}}\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, r, t)\cdot\tilde{\boldsymbol{r}}^{EMA}\big],$$

where $\tilde{\boldsymbol{r}}^{EMA} := V_{\boldsymbol{\theta}}^{EMA} - \boldsymbol{v}(\boldsymbol{x}_t, t)$ is the mean-field residual under the EMA tangent, and no $\boldsymbol{J}\boldsymbol{v}'$ noise term appears.

Proposition 5 (Bias-Variance Tradeoff)**.** *The EMA mean-field residual decomposes as*

$$\tilde{\boldsymbol{r}}^{EMA} \;=\; \boldsymbol{r}_{\boldsymbol{\theta}} \;+\; (t{-}r)\,\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t, t, t) - \boldsymbol{v}(\boldsymbol{x}_t, t)),$$

where $\boldsymbol{r}_{\boldsymbol{\theta}}$ is the true MeanFlow residual from equation 6. At convergence $\boldsymbol{r}_{\boldsymbol{\theta}} \to \boldsymbol{0}$ and the FM anchor drives $\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t, t, t) \to \boldsymbol{v}(\boldsymbol{x}_t, t)$, giving $\tilde{\boldsymbol{r}}^{EMA} \to \boldsymbol{0}$.

Together, proposition 4 eliminates the $\boldsymbol{J}\boldsymbol{v}'$ noise (cf. theorem 2), and proposition 5 bounds the transient EMA bias by $(t{-}r)\|\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}}\|\,\|\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t, t, t) - \boldsymbol{v}(\boldsymbol{x}_t, t)\|$, which the FM anchor drives to zero.

3.2.3 Adaptive Weights

Curvature gap and flow acceleration. Although the EMA tangent removes the Jacobian-amplified variance from the *gradient*, samples with large $\|\boldsymbol{J}\|$ still produce loss values that dwarf samples in flat, low-curvature regions, biasing the empirical-risk estimator toward the noisy regime. The size of $\|\boldsymbol{J}\|$ is in turn controlled by the curvature gap $\Delta(\boldsymbol{x}_t, r, t) := \boldsymbol{u}(\boldsymbol{x}_t, r, t) - \boldsymbol{v}(\boldsymbol{x}_t, t)$, whose magnitude admits a closed-form Taylor expansion along the flow.

Theorem 6 (Flow Acceleration Expansion of the Curvature Gap)**.** *Let $\boldsymbol{v}(\boldsymbol{x}, t) \in C^2$. The curvature gap admits a Taylor expansion at $(\boldsymbol{x}_t, t)$ along the flow:*

$$\Delta(\boldsymbol{x}_t, r, t) \;=\; -\frac{(t{-}r)}{2}\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}(\boldsymbol{x}_t, t) + \mathcal{R}_2(\boldsymbol{x}_t, r, t),$$

where $\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t} \triangleq \partial_t\boldsymbol{v} + (\nabla_{\boldsymbol{x}}\boldsymbol{v})\,\boldsymbol{v}$ is the material derivative (a.k.a. flow acceleration) and $\mathcal{R}_2 = \mathcal{O}((t{-}r)^2)$.

The C^2 assumption is benign because existing backbones [29, 12] use twice-differentiable activations (e.g., GELU [30]); proof in Appendix C. Theorem 6 clarifies prior heuristics: rectified flows [31] drive $\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t} \to \boldsymbol{0}$ to eliminate Δ, and progressive scheduling [26] shrinks the prefactor by training with small $r \to t$. The growth factor $\frac{t-r}{2}\|\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}\|$ is the principled criterion both heuristics target, and motivates a single per-sample weight that downweights high-curvature, long-interval samples directly.

We derive the adaptive weight using the Gauss-Markov Theorem. In OT-CFM, we argue it is fair to treat the conditional fluctuation $\boldsymbol{v}'$ as Gaussian noise, and derive the Best Linear Unbiased Estimator (BLUE) of the mean-field residual $\tilde{\boldsymbol{r}}^{\text{EMA}}$ with the inverse-covariance matrix weight $(\boldsymbol{J}\Sigma_{\boldsymbol{v}'}\boldsymbol{J}^\top)^{-1}$. However, it further introduce two practical pathologies. First, the per-sample $d \times d$ matrix inversion has a cubic-cost $\mathcal{O}(d^3)$ and is intractable for high-dimensional data or latents. Meanwhile, the matrix is ill-conditioned whenever $(t{-}r)\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}} \to \boldsymbol{I}_d$ and $\boldsymbol{J} \approx \boldsymbol{0}$, which pushes the inverse to $+\infty$.

To address these pathologies, we propose to use isotropic-noise approximation $\Sigma_{\boldsymbol{v}'} \approx \sigma_t^2\boldsymbol{I}_d$. Because $\mathrm{Tr}(\boldsymbol{J}\Sigma_{\boldsymbol{v}'}\boldsymbol{J}^\top) = \sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)$, we define the per-sample *trace weight*

Algorithm 1 Variance-Aware MeanFlow Training

Require: Dataset $\mathcal{D}$, EMA decay μ, anchor weight λ, anchor interval $[\delta_{\min}, \delta_{\max}]$
Require: Schedule σ_t, number of probes K
  Initialize $\bar{\boldsymbol{\theta}} \leftarrow \boldsymbol{\theta}$
  for $k = 1, 2, \dots$ **do**
    Sample $(\boldsymbol{x}_0, \boldsymbol{x}_1) \sim \mathcal{D} \times \mathcal{N}(\mathbf{0}, \boldsymbol{I}_d),\ t \sim \mathcal{U}[0,1],\ r \sim \mathcal{U}[0,t],\ \delta \sim \mathcal{U}[\delta_{\min}, \delta_{\max}]$
    $\boldsymbol{x}_t \leftarrow (1-t)\boldsymbol{x}_0 + t\,\boldsymbol{x}_1, \quad \boldsymbol{v}_{\text{cond}} \leftarrow \boldsymbol{x}_1 - \boldsymbol{x}_0$
    $\boldsymbol{v}_{\text{tang}} \leftarrow \mathtt{sg}[\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t, t, t)]$ ▷ EMA tangent
    $V_{\boldsymbol{\theta}} \leftarrow \boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, r, t) + (t-r)\,\mathtt{sg}[\mathtt{JVP}(\boldsymbol{u}_{\boldsymbol{\theta}}, (\boldsymbol{x}_t, r, t), (\boldsymbol{v}_{\text{tang}}, 0, 1))]$
    Sample K probes $\{\boldsymbol{z}_i\}_{i=1}^{K} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I}_d)$
    $\widehat{\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)} \leftarrow \frac{1}{K}\sum_{i=1}^{K} \big\|(t-r)\,\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, r, t)\,\boldsymbol{z}_i - \boldsymbol{z}_i\big\|_2^2$
    $w \leftarrow 1 \,/\, \big(1 + \sigma_t^2\,\mathtt{sg}[\widehat{\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)}]/d\big)$ ▷ trace weight
    $\ell_{\text{MF}} \leftarrow w \cdot \|V_{\boldsymbol{\theta}} - \boldsymbol{v}_{\text{cond}}\|_2^2$
    $\ell_{\text{FM}} \leftarrow \|\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, t-\delta, t) - \boldsymbol{v}_{\text{cond}}\|_2^2$ ▷ FM anchor
    $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta\,\nabla_{\boldsymbol{\theta}}(\ell_{\text{MF}} + \lambda\,\ell_{\text{FM}})$
    $\bar{\boldsymbol{\theta}} \leftarrow \mu\bar{\boldsymbol{\theta}} + (1-\mu)\boldsymbol{\theta}$ ▷ EMA update
  end for
  return $\boldsymbol{\theta}$ ▷ Generate: $\hat{\boldsymbol{x}}_0 = \boldsymbol{x}_1 - \boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}_1, 0, 1)$

$$w(\boldsymbol{x}_t, r, t) \;:=\; \frac{1}{1 + \sigma_t^2\,\mathtt{sg}[\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)]\,/d} \;=\; \frac{d}{d + \sigma_t^2\,\mathtt{sg}[\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)]}, \tag{11}$$

where d is the data dimensionality. Appendix C.6 derives this form as a Tikhonov-regularized BLUE-scalar approximation under $\Sigma_{\boldsymbol{v}'} \approx \sigma_t^2\boldsymbol{I}_d$, in which the additive d caps the per-sample noise contribution at the data dimensionality. Still, the direct evaluation of $\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)$ requires d Jacobian-vector products and is intractable for high-dimensional data. We instead use a single Rademacher probe $\boldsymbol{z} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I}_d)$ and the unbiased Hutchinson's estimator [32]

$$\widehat{\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)} \;=\; \big\|\boldsymbol{J}\boldsymbol{z}\big\|_2^2 \;=\; \big\|(t-r)\,\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}}\,\boldsymbol{z} - \boldsymbol{z}\big\|_2^2, \tag{12}$$

which costs a single JVP per step. Multiple probes can be averaged when extra accuracy is needed. We find $K = 1$ is sufficient for our experiment with DiT on ImageNet dataset.

In addition, the scalar σ_t in equation 11 is the *time-dependent* standard-deviation scale of the isotropic-noise approximation $\Sigma_{\boldsymbol{v}'} \approx \sigma_t^2\boldsymbol{I}_d$; the actual conditional-velocity variance depends on p_{data} and is generally not closed-form. Under Laplace's approximation $p_{\text{data}} \approx \mathcal{N}(\mathbf{0}, \sigma_d^2\boldsymbol{I}_d)$ [33, 34], we use conjugate-Gaussian inference to derive

$$\Sigma_{\boldsymbol{v}'}(\boldsymbol{x}_t, t) \;=\; \frac{\sigma_d^2}{\sigma_d^2(1-t)^2 + t^2}\,\boldsymbol{I}_d, \tag{13}$$

which is U-shaped in t and peaks near $t = 0.5$ where conditional paths most strongly intersect—the same regime where the Jacobian-deviation norm peaks in theorem 2. Nevertheless, p_{data} in practice is generally non-Gaussian, we thereby treat σ_t as a hyperparameter and sweep five candidates in our experiments. See derivation and additional results in appendix D.2.

Proposition 7 (Trace-Weight Properties)**.** *Under the trace-weight loss equation 10 and the isotropic approximation $\Sigma_{\boldsymbol{v}'} \approx \sigma_t^2\boldsymbol{I}_d$: **(a)** the stop-gradient on $\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)$ keeps w out of the parameter update, so the gradient inherits the structure of proposition 4 unchanged; **(b)** the expected per-sample noise contribution $w\,\sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top) = d\,\sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)/(d + \sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)) \le d$ is uniformly capped at the data dimensionality, eliminating the unbounded curvature growth from theorem 2; **(c)** w is strictly decreasing in $\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)$ and, by theorem 6, in $(t-r)^2\|\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}\|^2$, so high-curvature long-interval samples are automatically downweighted—subsuming inverse-variance weighting, progressive scheduling, and acceleration-aware interval sampling in a single per-sample scalar.*

The full training procedure is in Algorithm 1 and the proof of proposition 7 in Appendix C; the connection between w and the acceleration-to-noise ratio $\mathrm{ANR} = (t-r)\|\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}\|/\sqrt{\mathrm{Tr}(\Sigma_{\boldsymbol{v}'})}$ is in Appendix D.

4 Related Work

MeanFlow and Variants. MeanFlow [24] learns the average velocity field via a total-derivative identity, enabling distillation-free one-step generation. AlphaFlow [25] identifies gradient conflict between the flow matching and total consistency components, proposing α-scheduling to manage the tradeoff. Improved MeanFlow [26] traces instability to the stochastic JVP tangent and introduces a separate velocity head with stop-gradient. Re-MeanFlow [35] addresses trajectory curvature through a reflow pre-processing step. Terminal Velocity Matching (TVM) [36] sidesteps the spatial Jacobian entirely by differentiating w.r.t. the terminal time, and Functional Mean Flows [37] independently establishes the marginal-conditional gap for average velocities in a Hilbert space setting. Physics-informed distillation [38] derives a related PDE-based loss for learning flow maps. Our curvature-variance principle unifies these diagnostics—our method addresses all failure modes without architectural changes, separate heads, or pre-training.

Variance Reduction in Flow Training. Several concurrent works address variance in flow-based training. Stable Velocity [39] reveals a two-regime variance structure in flow matching and proposes composite conditioning; TPC [40] reduces variance through temporal pair coupling; and preconditioned flow matching [41] addresses ill-conditioning via anisotropic preconditioning. Bertrand et al. [42] argue that stochastic target variance is negligible for standard flow matching—consistent with our analysis, since the problematic amplification is specific to the MeanFlow JVP, not the flow matching loss itself. Rectified flows [31, 43] reduce trajectory curvature by iterative straightening, which by theorem 6 drives $\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t} \to 0$.

5 Experiments

We evaluate VaMF on two fronts: **(i)** a 2-D toy benchmark spanning four standard distributions, where we measure both sliced Wasserstein-1 and the variance-amplification diagnostics that motivate the trace weight, and **(ii)** a high-dimensional dense-Gaussian-mixture (DGMM) scaling study in which we sweep the data dimensionality from $d{=}2$ up to $d{=}64$. Section 5.4 ablates the trace-weight component and the schedule σ_t in equation 11; a class-conditional DiT-B/4 latent-space scale-up on ImageNet-256 is reported in Appendix E. All toy and DGMM experiments use a 3-layer MLP with 128 hidden units, $200{,}000$ optimization steps, and three random seeds $\{42, 0, 1\}$. Code, configurations, and result JSONs are released with the paper.

5.1 Variance Amplification and the Trace Term

We first probe the per-sample loss variance and Jacobian-deviation norm on a frozen checkpoint trained on the eight-Gaussians dataset. Figure 2 plots the variance ratio $\mathrm{Var}[\mathcal{L}_{\text{stoch}}]/\mathrm{Var}[\mathcal{L}_{\text{det}}]$ as a function of t (panel a) and, on a shared t-axis, the per-sample loss variance $\mathrm{Var}[\mathcal{L}_{\text{det}}]$ alongside the Frobenius norm $\|(t{-}r)J - I\|_F$ that drives the amplification (panel b). The variance ratio peaks at $\sim 259\times$ near $t \approx 0.4$, in direct empirical agreement with theorem 2; the per-sample loss variance does not vanish anywhere on $t \in [0, 1]$ and its scale tracks $\|(t{-}r)J - I\|_F$ across the integration window, identifying the Jacobi factor as the immediate source of the amplification.

5.2 Toy Benchmark

Table 1 reports both SW_1 and SW_2 on the four 2-D toy datasets, averaged over three random seeds. VaMF-TW achieves the best score on every dataset under both metrics, with relative SW_1 improvements of -23.4% on `checkerboard`, -13.3% on `eight_gaussians`, -22.6% on `two_moons`, and -54.0% on `swiss_roll`. The SW_2 improvements are systematically larger (-25.2%, -3.6%, -47.5%, and -66.0% respectively): the squared-distance metric weights configurations with isolated outliers more heavily, so the larger SW_2 gains are direct evidence that VaMF-TW reduces the per-sample tail of the residual distribution—exactly the variance-amplification regime that the trace weight was designed to suppress. The qualitative picture is consistent with the SW numbers: Figure 3 shows the full samples grid, where VaMF-TW recovers structural details—spirals, sharp corners, mode separation—that vanilla MeanFlow blurs out.

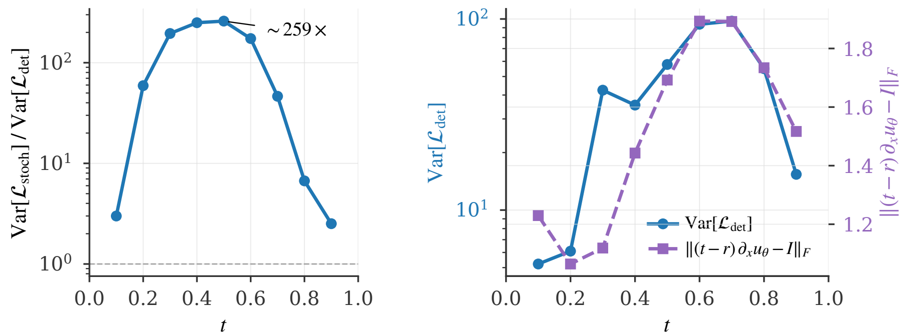

Figure 2: Variance amplification in vanilla MeanFlow. **(a)** The stochastic-vs-deterministic variance ratio $\mathrm{Var}[\mathcal{L}_{\mathrm{stoch}}]/\mathrm{Var}[\mathcal{L}_{\mathrm{det}}]$ peaks at $\sim 259\times$ near $t \approx 0.4$, in direct empirical agreement with theorem 2. **(b)** The per-sample loss variance $\mathrm{Var}[\mathcal{L}_{\mathrm{det}}]$ (left axis) is non-vanishing across t and co-varies with the Jacobi-factor norm $\|(t-r)J - I\|_F$ (right axis): both quantities peak in the same mid-t regime, empirically certifying that the Frobenius norm of $(t-r)J - I$ governs the per-sample loss scale.

Table 1: Sliced Wasserstein distance (lower is better) on the 2-D toy benchmark, averaged over three seeds $\{42, 0, 1\}$ at 200k steps. SW_p is the sliced Wasserstein-p distance evaluated on 4096 samples with 500 random projections; both metrics use the *same* projection key per evaluation for variance-reduced paired estimates. VaMF-TW uses $\sigma_t = 1$ (the schedule-sweep winner from Figure 5) and a single Hutchinson probe.

Dataset	Metric	MeanFlow	VaMF-TW
`checkerboard`	SW_1	0.175 ± 0.036	$\mathbf{0.134 \pm 0.029}$
	SW_2	0.234 ± 0.042	$\mathbf{0.175 \pm 0.046}$
`eight_gaussians`	SW_1	0.098 ± 0.016	$\mathbf{0.085 \pm 0.010}$
	SW_2	0.151 ± 0.022	$\mathbf{0.146 \pm 0.023}$
`two_moons`	SW_1	0.093 ± 0.028	$\mathbf{0.072 \pm 0.019}$
	SW_2	0.177 ± 0.068	$\mathbf{0.093 \pm 0.025}$
`swiss_roll`	SW_1	0.098 ± 0.058	$\mathbf{0.045 \pm 0.003}$
	SW_2	0.177 ± 0.151	$\mathbf{0.060 \pm 0.007}$

5.3 High-Dimensional Scaling

We sweep the data dimensionality on the dense Gaussian mixture (DGMM) defined in our toy framework—64 overlapping isotropic components on a unit sphere of radius 1.5 in $\mathbb{R}^d$, $d \in \{2, 4, 8, 16, 32, 64\}$—which is constructed so that variance amplification dominates the optimization landscape. Table 2 reports SW_1 at convergence (seed 42, single run per cell). VaMF-TW is the best or tied-best for every dimension $d \geq 4$, and the gap to MeanFlow widens with d (e.g. -16.5% at $d{=}64$ versus $+20.1\%$ at $d{=}2$, where the curvature term is too small to matter), in line with the prediction of theorem 2 that the amplification term scales with $\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)$ and hence with d.

5.4 Ablation Studies

Component ablation: is the trace weight load-bearing? VaMF-L_2 keeps the EMA tangent and FM anchor (Sections 3.2.1 and 3.2.2) but drops the trace weight; comparing it against vanilla MeanFlow and VaMF-TW isolates the contribution of the curvature-aware reweighting (Section 3.2.3). On the toy benchmark, VaMF-L_2 recovers a partial improvement over MeanFlow but trails VaMF-TW on every dataset (Table 3); the gap is largest on `checkerboard` (-19% trace weight vs MeanFlow, -6% for VaMF-L_2) and `two_moons` (-22% vs -6%). The same picture appears dynamically in Figure 4: vanilla MeanFlow plateaus with persistent high-frequency oscillations driven by the gradient variance of theorem 2; VaMF-L_2 converges smoothly but to a higher steady-state SW_1 than VaMF-TW on every dataset. The trace weight is the load-bearing piece.

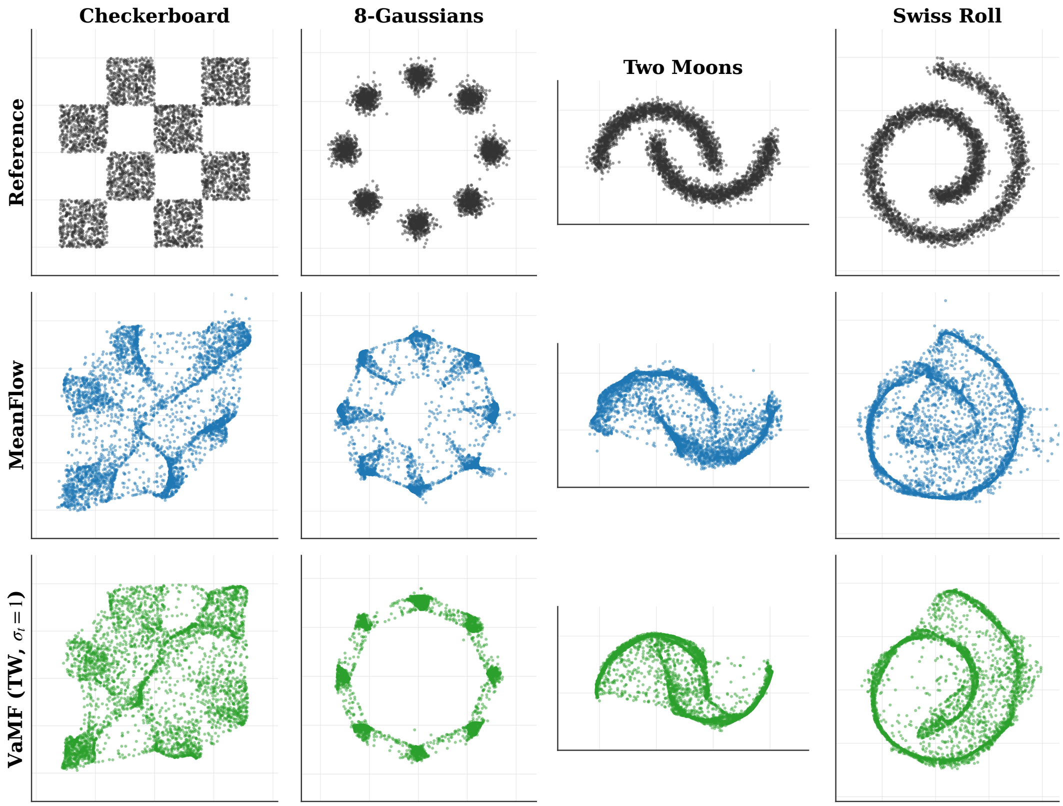

Figure 3: Reference distribution and final-step samples on the four 2-D toy datasets at seed 42. VaMF-TW (bottom row) recovers the structural details—spirals, sharp corners, mode separation—that vanilla MeanFlow blurs out, especially on `checkerboard` and `swiss_roll`.

Table 2: DGMM scaling (single seed, 200k steps): SW_1 at convergence as the dimension d grows. The trace weight's advantage over MeanFlow widens with d.

d	2	4	8	16	32	64
MeanFlow	**0.097**	0.091	0.078	**0.050**	0.035	0.036
VaMF-TW	0.117	**0.089**	**0.077**	0.051	**0.035**	**0.030**

Schedule sweep on toy benchmark. Figure 5 compares the five σ_t schedules of Appendix D.2—NONE ($\sigma_t = 1$), $\sigma_t = t$, $\sigma_t = t^2$, the U-shape $\sigma_t = \sqrt{t(1-t)}$, and the BLUE under Laplace's approximation of p_{data}—across the four toy datasets. The bare BLUE-scalar form $\sigma_t = 1$ delivers the best mean SW_1 (0.084) and ties the U-shape on `swiss_roll`; the monotone schedules ($\sigma_t = t$, $\sigma_t = t^2$) trail by 4–6%, and the BLUE-GAUSS schedule is consistently worst (the data here is multimodal, so the unimodal-Gaussian Laplace approximation is loose). We therefore adopt $\sigma_t = 1$ as the default for all toy and DiT experiments.

Schedule ablation on DGMM. Table 4 repeats the schedule ablation in the high-dimensional regime where the data is approximately isotropic Gaussian. There $\sigma_t = t^2$ wins at d=2 (where time-axis attenuation matters most) and is tied at higher d; the learned MLP variant tracks the quadratic schedule but offers no consistent improvement. Combined with the toy sweep, we recommend $\sigma_t = 1$ for non-Gaussian data and $\sigma_t = t^2$ when the data is approximately isotropic Gaussian.

5.5 Discussion

The toy and DGMM experiments are direct empirical instances of the curvature-variance principle: VaMF-TW dominates exactly where the trace term $\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)$ is large enough to matter (high-

Table 3: Component ablation (toy benchmark, mean SW_1 over $\{42, 0, 1\}$ at 200k steps). VaMF-L_2 keeps the EMA tangent and FM anchor but drops the trace weight; it captures part of the gain but trails VaMF-TW everywhere, confirming the trace weight is load-bearing.

Method	`checkerboard`	`eight_gaussians`	`two_moons`	`swiss_roll`
MeanFlow	0.175	0.098	0.093	0.098
VaMF-L_2	0.165	0.087	0.087	0.048
VaMF-TW	**0.134**	**0.085**	**0.072**	**0.045**

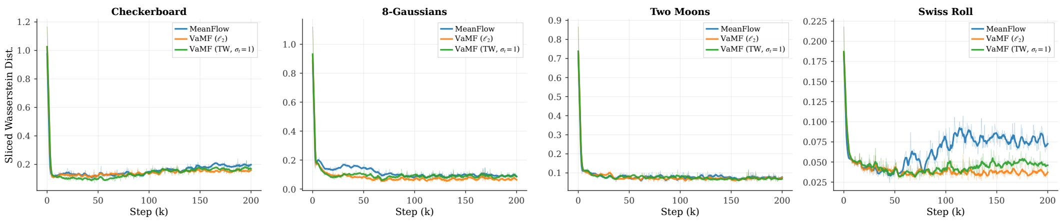

Figure 4: Convergence stability across the 2-D toy benchmark: SW_1 vs training step (smoothed) for each dataset. Vanilla MeanFlow plateaus with persistent high-frequency oscillations driven by the per-step gradient variance of theorem 2; VaMF-L_2 converges smoothly but to a higher floor than VaMF-TW with $\sigma_t = 1$, and the gap between VaMF-L_2 and VaMF-TW is what the trace weight contributes.

curvature regions of swiss_roll, high-d DGMM, and the variance-amplified middle of the integration window in Section 5.1), and reduces to the baseline behavior in low-curvature regimes (e.g. low-d DGMM at d=2). The same mechanism transfers from d=2 toys to d=4096 image latents without architectural modification: a class-conditional DiT-B/4 / ImageNet-256 latent-space training shows VaMF-TW maintains a small but consistent training-loss improvement over the matched MeanFlow baseline at a $\sim$22% wall-clock overhead per step (Appendix E).

6 Conclusion

We introduced the **curvature-variance principle**, a unifying theoretical lens that attributes the three independently reported MeanFlow training pathologies—gradient conflict [25], Jacobian amplification [26], and the curvature bottleneck [35]—to a single quantity: the material derivative of the marginal velocity field. The principle yields three design directives, which we instantiate as **Variance-Aware MeanFlow (VaMF)**—an EMA velocity anchor (Section 3.2.1) that eliminates the Jacobian-amplified gradient noise, a flow-matching anchor (Section 3.2.2) that supervises the boundary condition without an additional head, and a closed-form Hutchinson-estimated trace weight (Section 3.2.3) that approximates the per-sample BLUE under isotropic conditional-velocity noise. VaMF preserves the original MeanFlow MSE objective and backbone, adds a single Hutchinson JVP per step, and consistently improves over vanilla MeanFlow on every dataset of our 2-D toy benchmark under both SW_1 and SW_2 (mean SW_1 improvement -28% across the four datasets) and dominates from d=4 to d=64 on the dense-Gaussian-mixture scaling sweep; Appendix E additionally shows the same mechanism transfers to DiT-B/4 / ImageNet-256 latent training at a $\sim$ 22% wall-clock overhead.

Limitations. The trace-weight derivation assumes isotropic conditional-velocity noise $\Sigma_{\boldsymbol{v}'} \approx \sigma_t^2 \boldsymbol{I}_d$; distributions with strongly correlated $\Sigma_{\boldsymbol{v}'}$ may benefit from a low-rank Jacobian factorization that we leave to future work. Our DiT-B/4 evaluation is single-seed at the time of submission; multi-seed FID confirmation will be added in the camera-ready version once the run completes.

Future work. Two natural extensions stand out. *(i) Anisotropic trace weights* learned from a low-rank approximation of $\Sigma_{\boldsymbol{v}'}$ would tighten the BLUE bound at the cost of a small auxiliary head—bridging the closed-form trace weight of this paper with the per-pixel variance head sketched in concurrent work. *(ii) Schedule design.* Our BLUE-GAUSS schedule for σ_t is principled only under

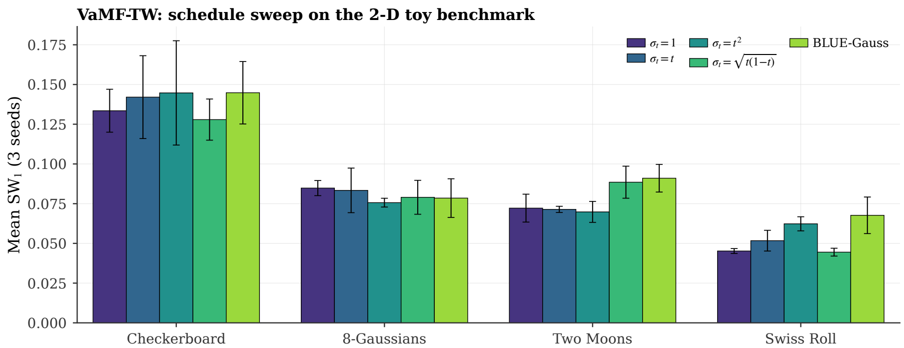

Figure 5: σ_t schedule sweep on the 2-D toy benchmark (mean SW_1 over 3 seeds, error bars ± 1 standard error). The bare BLUE-scalar $\sigma_t = 1$ has the best per-dataset average and is parameter-free, motivating its choice as the VaMF-TW default. BLUE-GAUSS underperforms because the toy data is multimodal, so the unimodal-Gaussian Laplace approximation of p_{data} is loose.

Table 4: Schedule ablation for VaMF-TW on DGMM (single seed, 200k steps): SW_1 as a function of σ_t. The quadratic schedule wins at low d where time-axis attenuation matters; the learned MLP variant offers no consistent gain.

d	2	4	8	16	32	64
$\sigma_t = 1$	0.113	**0.089**	**0.077**	0.052	**0.035**	0.032
$\sigma_t = t^2$	**0.087**	0.092	0.078	**0.049**	**0.035**	**0.030**
$\sigma_t = \mathtt{NN}_{\boldsymbol{\phi}}(t)$	0.119	**0.089**	0.079	0.051	**0.035**	0.031

Laplace's approximation of p_{data}; more general data distributions may benefit from a learned or moment-matched alternative that directly tracks $\mathrm{Tr}(\Sigma_{\boldsymbol{v}'})/d$.

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References

[1] Ricky T. Q. Chen, Yulia Rubanova, Jesse Bettencourt, and David K. Duvenaud. Neural ordinary differential equations. In *Advances in Neural Information Processing Systems*, volume 31, 2018.

[2] Danilo Rezende and Shakir Mohamed. Variational inference with normalizing flows. In *Proceedings of the 32nd International Conference on Machine Learning*, volume 37 of *Proceedings of Machine Learning Research*, pages 1530–1538, 2015.

[3] Laurent Dinh, Jascha Sohl-Dickstein, and Samy Bengio. Density estimation using Real-NVP. In *International Conference on Learning Representations*, 2017.

[4] Will Grathwohl, Ricky T. Q. Chen, Jesse Bettencourt, Ilya Sutskever, and David Duvenaud. FFJORD: Free-form continuous dynamics for scalable reversible generative models. In *International Conference on Learning Representations*, 2019.

[5] Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matt Le. Flow matching for generative modeling, 2023.

[6] Alexander Tong, Kilian Fatras, Nikolay Malkin, Guillaume Huguet, Yanlei Zhang, Jarrid Rector-Brooks, Guy Wolf, and Yoshua Bengio. Improving and generalizing flow-based generative models with minibatch optimal transport, 2024.

[7] Michael S. Albergo and Eric Vanden-Eijnden. Building normalizing flows with stochastic interpolants, 2023.

[8] Michael Albergo, Nicholas M. Boffi, and Eric Vanden-Eijnden. Stochastic interpolants: A unifying framework for flows and diffusions. *Journal of Machine Learning Research*, 26(209):1–80, 2025.

[9] Jascha Sohl-Dickstein, Eric Weiss, Niru Maheswaranathan, and Surya Ganguli. Deep unsupervised learning using nonequilibrium thermodynamics. In *Proceedings of the 32nd International Conference on Machine Learning*, volume 37 of *Proceedings of Machine Learning Research*, pages 2256–2265, 2015.

[10] Yang Song and Stefano Ermon. Generative modeling by estimating gradients of the data distribution. In *Advances in Neural Information Processing Systems*, volume 32, 2019.

[11] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In H. Larochelle, M. Ranzato, R. Hadsell, M.F. Balcan, and H. Lin, editors, *Advances in Neural Information Processing Systems*, volume 33, pages 6840–6851. Curran Associates, Inc., 2020.

[12] Alexander Quinn Nichol and Prafulla Dhariwal. Improved denoising diffusion probabilistic models. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 8162–8171, 2021.

[13] Yang Song, Jascha Sohl-Dickstein, Diederik P. Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. Score-based generative modeling through stochastic differential equations. In *International Conference on Learning Representations*, 2021.

[14] Prafulla Dhariwal and Alexander Nichol. Diffusion models beat GANs on image synthesis. In *Advances in Neural Information Processing Systems*, volume 34, 2021.

[15] Tero Karras, Miika Aittala, Timo Aila, and Samuli Laine. Elucidating the design space of diffusion-based generative models. In *Advances in Neural Information Processing Systems*, volume 35, 2022.

[16] Diederik P. Kingma and Max Welling. Auto-encoding variational Bayes. In *International Conference on Learning Representations*, 2014.

[17] Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models. In *International Conference on Learning Representations*, 2021.

[18] Tim Salimans and Jonathan Ho. Progressive distillation for fast sampling of diffusion models, 2022.

[19] Tianwei Yin, Michaël Gharbi, Richard Zhang, Eli Shechtman, Frédo Durand, William T. Freeman, and Taesung Park. One-step diffusion with distribution matching distillation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 6613–6623, June 2024.

[20] Tianwei Yin, Michaël Gharbi, Taesung Park, Richard Zhang, Eli Shechtman, Frédo Durand, and William T. Freeman. Improved distribution matching distillation for fast image synthesis. In A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. Tomczak, and C. Zhang, editors, *Advances in Neural Information Processing Systems*, volume 37, pages 47455–47487. Curran Associates, Inc., 2024.

[21] Yang Song, Prafulla Dhariwal, Mark Chen, and Ilya Sutskever. Consistency models. In *Proceedings of the 40th International Conference on Machine Learning*, ICML'23. JMLR.org, 2023.

[22] Cheng Lu and Yang Song. Simplifying, stabilizing and scaling continuous-time consistency models, 2025.

[23] Nicholas M. Boffi, Michael S. Albergo, and Eric Vanden-Eijnden. Flow map matching with stochastic interpolants: A mathematical framework for consistency models, 2025.

[24] Zhengyang Geng, Mingyang Deng, Xingjian Bai, J. Zico Kolter, and Kaiming He. Mean flows for one-step generative modeling, 2025.

[25] Huijie Zhang, Aliaksandr Siarohin, Willi Menapace, Michael Vasilkovsky, Sergey Tulyakov, Qing Qu, and Ivan Skorokhodov. Alphaflow: Understanding and improving meanflow models, 2025.

[26] Zhengyang Geng, Yiyang Lu, Zongze Wu, Eli Shechtman, J. Zico Kolter, and Kaiming He. Improved mean flows: On the challenges of fastforward generative models, 2025.

[27] Boris T. Polyak and Anatoli B. Juditsky. Acceleration of stochastic approximation by averaging. volume 30, pages 838–855, 1992.

[28] Antti Tarvainen and Harri Valpola. Mean teachers are better role models: Weight-averaged consistency targets improve semi-supervised learning results. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

[29] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-net: Convolutional networks for biomedical image segmentation. In *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015*, pages 234–241, 2015.

[30] Dan Hendrycks and Kevin Gimpel. Gaussian error linear units (GELUs), 2016.

[31] Xingchao Liu, Chengyue Gong, and Qiang Liu. Flow straight and fast: Learning to generate and transfer data with rectified flow, 2022.

[32] M. F. Hutchinson. A stochastic estimator of the trace of the influence matrix for Laplacian smoothing splines. *Communications in Statistics - Simulation and Computation*, 18(3):1059–1076, 1989.

[33] Luke Tierney and Joseph B. Kadane. Accurate approximations for posterior moments and marginal densities. *Journal of the American Statistical Association*, 81(393):82–86, 1986.

[34] David J. C. MacKay. A practical Bayesian framework for backpropagation networks. *Neural Computation*, 4(3):448–472, 1992.

[35] Xinxi Zhang, Shiwei Tan, Quang Nguyen, Quan Dao, Ligong Han, Xiaoxiao He, Tunyu Zhang, Chengzhi Mao, Dimitris Metaxas, and Vladimir Pavlovic. Overcoming the curvature bottleneck in meanflow, 2026.

[36] Linqi Zhou, Mathias Parger, Ayaan Haque, and Jiaming Song. Terminal velocity matching, 2026.

[37] Zhiqi Li, Yuchen Sun, Greg Turk, and Bo Zhu. Functional mean flow in hilbert space, 2025.

[38] Joshua Tian Jin Tee, Kang Zhang, Hee Suk Yoon, Dhananjaya Nagaraja Gowda, Chanwoo Kim, and Chang D. Yoo. Physics informed distillation for diffusion models, 2024.

[39] Donglin Yang, Yongxing Zhang, Xin Yu, Liang Hou, Xin Tao, Pengfei Wan, Xiaojuan Qi, and Renjie Liao. Stable velocity: A variance perspective on flow matching, 2026.

[40] Chika Maduabuchi and Jindong Wang. Temporal pair consistency for variance-reduced flow matching, 2026.

[41] Shadab Ahamed, Eshed Gal, Simon Ghyselincks, Md Shahriar Rahim Siddiqui, Moshe Eliasof, and Eldad Haber. Preconditioned score and flow matching, 2026.

[42] Quentin Bertrand, Anne Gagneux, Mathurin Massias, and Rémi Emonet. On the closed-form of flow matching: Generalization does not arise from target stochasticity, 2025.

[43] Sangyun Lee, Zinan Lin, and Giulia Fanti. Improving the training of rectified flows. In A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. Tomczak, and C. Zhang, editors, *Advances in Neural Information Processing Systems*, volume 37, pages 63082–63109. Curran Associates, Inc., 2024.

[44] William Peebles and Saining Xie. Scalable diffusion models with transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 4195–4205, 2023.

Supplementary Material

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A Notations

Table 5 lists all the notations we use in this work.

Table 5: Table of Notations.

Notation	Description
x, x_t	Random variable / random variable at time t
$\boldsymbol{x}$, $\boldsymbol{x}_t$	Realized state / state at time t: $\boldsymbol{x}_t = (1-t)\boldsymbol{x}_0 + t\boldsymbol{x}_1$
$\boldsymbol{v}(\mathrm{x}, t)$	Marginal velocity field
$\boldsymbol{v}_{\text{cond}}$	Conditional velocity: $\boldsymbol{v}_{\text{cond}} = \boldsymbol{x}_1 - \boldsymbol{x}_0$
$\boldsymbol{v}'$	Velocity fluctuation: $\boldsymbol{v}' = \boldsymbol{v}_{\text{cond}} - \boldsymbol{v}(\boldsymbol{x}_t, t)$, $\mathbb{E}_{\boldsymbol{x}_0\mid\boldsymbol{x}_t}[\boldsymbol{v}'] = \boldsymbol{0}$
$\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}, r, t)$	Learned average velocity field (two-parameter)
$\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}$	EMA average velocity with parameters $\bar{\boldsymbol{\theta}}$
$V_{\boldsymbol{\theta}}^{\text{EMA}}$	Compound prediction equation 8
$\boldsymbol{J}$	Jacobi factor: $\boldsymbol{J} = (t-r)\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}} - \boldsymbol{I}_d$
$\Sigma_{\boldsymbol{v}'}$	Conditional velocity covariance: $\mathrm{Cov}_{\boldsymbol{x}_0\mid\boldsymbol{x}_t}[\boldsymbol{v}']$
σ_t	Conditional-velocity variance scale (schedule, default $\sigma_t = 1$)
w	Trace weight: $w = 1/(1 + \sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)/d)$, see equation 11
$\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}$	Material derivative (flow acceleration): $\partial_t\boldsymbol{v} + (\nabla_{\boldsymbol{x}}\boldsymbol{v})\boldsymbol{v}$
$\Delta(\boldsymbol{x}_t, r, t)$	Curvature gap: $\boldsymbol{u}(\boldsymbol{x}_t, r, t) - \boldsymbol{v}(\boldsymbol{x}_t, t)$
$\mathtt{sg}[\cdot]$	Stop-gradient operator

B Proof of Lemma

Lemma 1. *Given vector fields $\boldsymbol{v}(\boldsymbol{x}, t \mid \boldsymbol{x}_0)$ that generate probability paths $p(\boldsymbol{x}, t \mid \boldsymbol{x}_0)$, for any data distribution $\mathrm{x}_0 \sim p(\mathrm{x}_0)$, the marginal velocity field is equal to the expected conditional velocity field*

$$\boldsymbol{v}(\boldsymbol{x}, t) = \mathbb{E}_{\boldsymbol{x}_0\sim p(\boldsymbol{x}_0, t\mid\boldsymbol{x})}\left[\boldsymbol{v}(\boldsymbol{x}, t \mid \boldsymbol{x}_0)\right].$$

Proof. A sufficient and necessary condition for a vector field to generate a probability path is the *continuity equation*, which is a partial differential equation (PDE) that writes

$$\frac{\partial}{\partial t}p(\boldsymbol{x}, t) + \mathrm{div}\left(p(\boldsymbol{x}, t)\boldsymbol{v}(\boldsymbol{x}, t)\right) = 0, \tag{14}$$

where $\mathrm{div}(f)$ denotes the divergence of function f. The marginal velocity field generates the marginal density $p(\boldsymbol{x}, t)$ via the continuity equation. By the law of total probability and the definition of the conditional velocity field:

$$p(\boldsymbol{x}, t)\,\boldsymbol{v}(\boldsymbol{x}, t) = \int p(\boldsymbol{x}, t \mid \boldsymbol{x}_0)\,\boldsymbol{v}(\boldsymbol{x}, t \mid \boldsymbol{x}_0)\,p(\boldsymbol{x}_0)\,\mathrm{d}\boldsymbol{x}_0, \tag{15}$$

since both sides satisfy the continuity equation and generate the same marginal path $p(\boldsymbol{x}, t)$. Dividing by $p(\boldsymbol{x}, t) = \int p(\boldsymbol{x}, t \mid \boldsymbol{x}_0)\,p(\boldsymbol{x}_0)\,\mathrm{d}\boldsymbol{x}_0$ and applying Bayes' rule $p(\boldsymbol{x}_0 \mid \boldsymbol{x}_t = \boldsymbol{x}) = p(\boldsymbol{x}, t \mid \boldsymbol{x}_0)\,p(\boldsymbol{x}_0)/p(\boldsymbol{x}, t)$:

$$\boldsymbol{v}(\boldsymbol{x}, t) = \int \boldsymbol{v}(\boldsymbol{x}, t \mid \boldsymbol{x}_0)\,p(\boldsymbol{x}_0 \mid \boldsymbol{x}_t = \boldsymbol{x})\,\mathrm{d}\boldsymbol{x}_0 = \mathbb{E}_{\mathrm{x}_0\mid\mathrm{x}_t=\boldsymbol{x}}[\boldsymbol{v}(\boldsymbol{x}, t \mid \mathrm{x}_0)]\,. \tag{16}$$

$\square$

C Proofs of Theorems and Propositions

C.1 Proof of Theorem 2 (Jacobian Variance Amplification)

Theorem 2 (Jacobian Variance Amplification)**.** *The Jacobi factor amplifies the gradient variance of the original MeanFlow loss:*

$$\mathrm{Var}[\nabla_{\boldsymbol{\theta}}\ell_{MF}(\boldsymbol{\theta}) \mid \boldsymbol{x}_t] \;\propto\; \mathrm{Tr}\big(\boldsymbol{J}\,\Sigma_{\boldsymbol{v}'}\,\boldsymbol{J}^\top\big)\,.$$

Proof. Under the stop-gradient operator, the compound prediction $V_{\boldsymbol{\theta}}^{\text{sg}}$ depends on $\boldsymbol{\theta}$ only through $\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, r, t)$. Let $\nabla_{\boldsymbol{\theta}} \boldsymbol{u}_{\boldsymbol{\theta}} \in \mathbb{R}^{d \times p}$ denote the parameter Jacobian of the network output. The per-step gradient is

$$\nabla_{\boldsymbol{\theta}} \ell_{\text{MF}} = 2\,(\nabla_{\boldsymbol{\theta}} \boldsymbol{u}_{\boldsymbol{\theta}})^{\top} (\boldsymbol{r}_{\boldsymbol{\theta}}^{\text{sg}} + \boldsymbol{J}\boldsymbol{v}')\,, \tag{17}$$

where $\boldsymbol{r}_{\boldsymbol{\theta}}^{\text{sg}}$ is the mean-field residual and $\boldsymbol{J} = (t-r)\partial_{\boldsymbol{z}} \boldsymbol{u}_{\boldsymbol{\theta}} - \boldsymbol{I}_d$ is the Jacobi factor. All three quantities—$\nabla_{\boldsymbol{\theta}} \boldsymbol{u}_{\boldsymbol{\theta}}$, $\boldsymbol{r}_{\boldsymbol{\theta}}^{\text{sg}}$, and $\boldsymbol{J}$—are deterministic conditioned on $\boldsymbol{x}_t$.

Since $\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}[\boldsymbol{v}'] = \mathbf{0}$ by equation 2, the conditional mean gradient is:

$$\mathbb{E}[\nabla_{\boldsymbol{\theta}} \ell \mid \boldsymbol{x}_t] = 2\,(\nabla_{\boldsymbol{\theta}} \boldsymbol{u}_{\boldsymbol{\theta}})^{\top}\, \boldsymbol{r}_{\boldsymbol{\theta}}^{\text{sg}}. \tag{18}$$

The centered gradient is $\nabla_{\boldsymbol{\theta}} \ell - \mathbb{E}[\nabla_{\boldsymbol{\theta}} \ell \mid \boldsymbol{x}_t] = 2(\nabla_{\boldsymbol{\theta}} \boldsymbol{u}_{\boldsymbol{\theta}})^{\top} \boldsymbol{J}\boldsymbol{v}'$, so the conditional variance is:

$$\begin{aligned}
\operatorname{Var}[\nabla_{\boldsymbol{\theta}} \ell \mid \boldsymbol{x}_t] &= \mathbb{E}\Big[\big\|2(\nabla_{\boldsymbol{\theta}} \boldsymbol{u}_{\boldsymbol{\theta}})^{\top} \boldsymbol{J}\boldsymbol{v}'\big\|^2 \mid \boldsymbol{x}_t\Big] \\
&= 4\,\mathbb{E}\left[(\boldsymbol{v}')^{\top} \boldsymbol{J}^{\top} \underbrace{(\nabla_{\boldsymbol{\theta}} \boldsymbol{u}_{\boldsymbol{\theta}})(\nabla_{\boldsymbol{\theta}} \boldsymbol{u}_{\boldsymbol{\theta}})^{\top}}_{\boldsymbol{G}_{\boldsymbol{\theta}} \succeq \mathbf{0}} \boldsymbol{J}\boldsymbol{v}' \;\middle|\; \boldsymbol{x}_t\right] \\
&= 4\,\operatorname{Tr}\big(\boldsymbol{G}_{\boldsymbol{\theta}}\, \boldsymbol{J}\, \Sigma_{\boldsymbol{v}'}\, \boldsymbol{J}^{\top}\big) \;\propto\; \operatorname{Tr}\big(\boldsymbol{J}\, \Sigma_{\boldsymbol{v}'}\, \boldsymbol{J}^{\top}\big)\,,
\end{aligned} \tag{19}$$

where $\boldsymbol{G}_{\boldsymbol{\theta}} = (\nabla_{\boldsymbol{\theta}} \boldsymbol{u}_{\boldsymbol{\theta}})(\nabla_{\boldsymbol{\theta}} \boldsymbol{u}_{\boldsymbol{\theta}})^{\top} \in \mathbb{R}^{d \times d}$ is the parameter-space Gram matrix and the last step uses $\boldsymbol{G}_{\boldsymbol{\theta}} \succeq \mathbf{0}$ to establish proportionality via the cyclic property of the trace. □

C.2 Proof of Theorem 3 (Semi-Gradient Gap)

Theorem 3 (Semi-Gradient Gap)**.** *The gradients of the MeanFlow loss with and without the stop-gradient operator differ by*

$$\underbrace{2(t-r)\,\mathbb{E}_{r,t,\boldsymbol{x}_0,\boldsymbol{x}_1}[\nabla_{\boldsymbol{\theta}}(\partial_{\boldsymbol{x}_t} \boldsymbol{u}_{\boldsymbol{\theta}} \cdot \boldsymbol{v}(\boldsymbol{x}_t, t) + \partial_t \boldsymbol{u}_{\boldsymbol{\theta}}) \cdot \boldsymbol{r}_{\boldsymbol{\theta}}]}_{\textit{mean-field gradient difference}} + \underbrace{\mathbb{E}_{r,t,\boldsymbol{x}_0,\boldsymbol{x}_1}\big[\nabla_{\boldsymbol{\theta}} \operatorname{Tr}\big(\boldsymbol{J}\, \Sigma_{\boldsymbol{v}'}\, \boldsymbol{J}^{\top}\big)\big]}_{\textit{variance-driven gradient difference}}.$$

Proof. The MeanFlow loss conditioned on $\boldsymbol{x}_t$ decomposes as (cf. equation 6):

$$\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}[\ell_{\text{MF}}] = \|\boldsymbol{r}_{\boldsymbol{\theta}}^{\text{sg}}\|^2 + \operatorname{Tr}\big(\boldsymbol{J}\, \Sigma_{\boldsymbol{v}'}\, \boldsymbol{J}^{\top}\big)\,. \tag{20}$$

Without stop-gradient. Both terms depend on $\boldsymbol{\theta}$ (through $\boldsymbol{u}_{\boldsymbol{\theta}}$ and $\partial_{\boldsymbol{z}} \boldsymbol{u}_{\boldsymbol{\theta}}$), so the full gradient is:

$$\nabla_{\boldsymbol{\theta}} \mathbb{E}[\ell] = \nabla_{\boldsymbol{\theta}}\, \|\boldsymbol{r}_{\boldsymbol{\theta}}\|^2 + \nabla_{\boldsymbol{\theta}} \operatorname{Tr}\big(\boldsymbol{J}\, \Sigma_{\boldsymbol{v}'}\, \boldsymbol{J}^{\top}\big)\,. \tag{21}$$

With stop-gradient. The JVP terms in $\boldsymbol{r}_{\boldsymbol{\theta}}^{\text{sg}}$ and $\boldsymbol{J}$ are treated as constants, so:

$$\nabla_{\boldsymbol{\theta}}^{\text{sg}}\, \|\boldsymbol{r}^{\text{sg}}\|^2 = 2\,(\nabla_{\boldsymbol{\theta}} \boldsymbol{u}_{\boldsymbol{\theta}})^{\top}\, \boldsymbol{r}^{\text{sg}}, \qquad \nabla_{\boldsymbol{\theta}}^{\text{sg}} \operatorname{Tr}\big(\boldsymbol{J}\, \Sigma_{\boldsymbol{v}'}\, \boldsymbol{J}^{\top}\big) = \mathbf{0}. \tag{22}$$

The gap between the full and stop-gradient gradients is therefore:

$$\begin{aligned}
&\nabla_{\boldsymbol{\theta}} \mathbb{E}[\ell] - \nabla_{\boldsymbol{\theta}}^{\text{sg}} \mathbb{E}[\ell] \\
&= \underbrace{\nabla_{\boldsymbol{\theta}}\, \|\boldsymbol{r}_{\boldsymbol{\theta}}\|^2 - 2\,(\nabla_{\boldsymbol{\theta}} \boldsymbol{u}_{\boldsymbol{\theta}})^{\top}\, \boldsymbol{r}^{\text{sg}}}_{\text{mean-field gradient difference}} + \underbrace{\nabla_{\boldsymbol{\theta}} \operatorname{Tr}\big(\boldsymbol{J}\, \Sigma_{\boldsymbol{v}'}\, \boldsymbol{J}^{\top}\big)}_{\text{variance-driven difference}}.
\end{aligned} \tag{23}$$

Expanding the first term: $\nabla_{\boldsymbol{\theta}} \|\boldsymbol{r}\|^2 = 2(\nabla_{\boldsymbol{\theta}} \boldsymbol{r}_{\boldsymbol{\theta}})^{\top} \boldsymbol{r}_{\boldsymbol{\theta}}$, and $\nabla_{\boldsymbol{\theta}} \boldsymbol{r}_{\boldsymbol{\theta}}$ includes the derivative of the JVP terms $(t-r)(\partial_{\boldsymbol{z}} \boldsymbol{u}_{\boldsymbol{\theta}} \cdot \boldsymbol{v} + \partial_t \boldsymbol{u}_{\boldsymbol{\theta}})$ w.r.t. $\boldsymbol{\theta}$, yielding the second-order mean-field difference $2(t-r)\mathbb{E}[\nabla_{\boldsymbol{\theta}}(\partial_{\boldsymbol{z}} \boldsymbol{u}_{\boldsymbol{\theta}} \cdot \boldsymbol{v} + \partial_t \boldsymbol{u}_{\boldsymbol{\theta}}) \cdot \boldsymbol{r}_{\boldsymbol{\theta}}]$. At convergence $\boldsymbol{r}_{\boldsymbol{\theta}} \to \mathbf{0}$, this term vanishes, and the gap is dominated by the variance-driven term $\nabla_{\boldsymbol{\theta}} \operatorname{Tr}(\boldsymbol{J}\, \Sigma_{\boldsymbol{v}'}\, \boldsymbol{J}^{\top})$. □

C.3 Proof of Theorem 6 (Flow Acceleration Expansion)

Theorem 4 (Flow Acceleration Expansion of the Curvature Gap)**.** *Let $\boldsymbol{v}(\boldsymbol{x}, t) \in C^2$. The curvature gap admits a Taylor expansion at $(\boldsymbol{x}_t, t)$ along the flow:*

$$\Delta(\boldsymbol{x}_t, r, t) \;=\; -\frac{(t-r)}{2}\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}(\boldsymbol{x}_t, t) + \mathcal{R}_2(\boldsymbol{x}_t, r, t),$$

where $\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t} \triangleq \partial_t \boldsymbol{v} + (\nabla_{\boldsymbol{x}} \boldsymbol{v})\, \boldsymbol{v}$ is the material derivative (a.k.a. flow acceleration) and $\mathcal{R}_2 = \mathcal{O}((t-r)^2)$.

Proof. Let $\boldsymbol{x}_\tau$ solve the ODE $\frac{\mathrm{d}\boldsymbol{x}_\tau}{\mathrm{d}\tau} = \boldsymbol{v}(\boldsymbol{x}_\tau, \tau)$ with terminal condition $\boldsymbol{x}_t = \boldsymbol{x}_t$. The curvature gap $\Delta = \boldsymbol{u}(\boldsymbol{x}_t, r, t) - \boldsymbol{v}(\boldsymbol{x}_t, t)$ can be written as:

$$\Delta(\boldsymbol{x}_t, r, t) = \frac{1}{t-r}\int_r^t \boldsymbol{v}(\boldsymbol{x}_\tau, \tau)\,\mathrm{d}\tau - \boldsymbol{v}(\boldsymbol{x}_t, t) = \frac{1}{t-r}\int_r^t [\boldsymbol{v}(\boldsymbol{x}_\tau, \tau) - \boldsymbol{v}(\boldsymbol{x}_t, t)]\,\mathrm{d}\tau. \tag{24}$$

Define $h = \tau - t \in [-(t-r), 0]$. Since $\boldsymbol{v} \in C^2$, we Taylor-expand $\boldsymbol{v}(\boldsymbol{x}_\tau, \tau)$ about $(\boldsymbol{x}_t, t)$ along the ODE trajectory:

$$\boldsymbol{v}(\boldsymbol{x}_\tau, \tau) = \boldsymbol{v}(\boldsymbol{x}_t, t) + \frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}(\boldsymbol{x}_t, t)\,h + \frac{1}{2}\frac{\mathrm{D}^2\boldsymbol{v}}{\mathrm{D}t^2}(\boldsymbol{x}_t, t)\,h^2 + O(h^3), \tag{25}$$

where $\frac{\mathrm{D}}{\mathrm{D}t}$ denotes the material derivative along the flow. Substituting and integrating over $h \in [-(t-r), 0]$:

$$\begin{aligned}
\Delta &= \frac{1}{t-r}\int_{-(t-r)}^{0}\left[\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}\,h + \frac{1}{2}\frac{\mathrm{D}^2\boldsymbol{v}}{\mathrm{D}t^2}\,h^2 + O(h^3)\right]\mathrm{d}h \\
&= \frac{1}{t-r}\left[\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}\cdot\frac{h^2}{2}\bigg|_{-(t-r)}^{0} + \frac{1}{2}\frac{\mathrm{D}^2\boldsymbol{v}}{\mathrm{D}t^2}\cdot\frac{h^3}{3}\bigg|_{-(t-r)}^{0} + O\big((t-r)^4\big)\right] \\
&= \frac{1}{t-r}\left[-\frac{(t-r)^2}{2}\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t} + \frac{(t-r)^3}{6}\frac{\mathrm{D}^2\boldsymbol{v}}{\mathrm{D}t^2} + O\big((t-r)^4\big)\right] \\
&= -\frac{(t-r)}{2}\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}(\boldsymbol{x}_t, t) + \frac{(t-r)^2}{6}\frac{\mathrm{D}^2\boldsymbol{v}}{\mathrm{D}t^2}(\boldsymbol{x}_t, t) + \mathcal{R}_3(\boldsymbol{x}_t, r, t),
\end{aligned} \tag{26}$$

where $\mathcal{R}_3 = O((t-r)^3)$ is the remainder. □

C.4 Proof of Proposition 4 (Gradient of VaMF Loss)

Proposition 5 (Gradient of VaMF Loss)**.** *The gradient of $\mathcal{L}_{MF}^{EMA}$ takes the form*

$$\nabla_{\boldsymbol{\theta}}\mathcal{L}_{MF}^{EMA} \;=\; 2\,\mathbb{E}\big[\nabla_{\boldsymbol{\theta}}\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, r, t)\cdot\tilde{\boldsymbol{r}}^{EMA}\big]\,,$$

where $\tilde{\boldsymbol{r}}^{EMA} := V_{\boldsymbol{\theta}}^{EMA} - \boldsymbol{v}(\boldsymbol{x}_t, t)$ is the mean-field residual under the EMA tangent, and no $\boldsymbol{J}\boldsymbol{v}'$ noise term appears.

Proof. The EMA velocity anchor $\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t, t, t)$ is deterministic given $\boldsymbol{x}_t$, since the EMA parameters $\bar{\boldsymbol{\theta}}$ are fixed during the gradient step. Under stop-gradient, $V_{\boldsymbol{\theta}}^{\mathrm{EMA}}$ depends on $\boldsymbol{\theta}$ only through $\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{x}_t, r, t)$. By the same argument as equation 17, the per-step gradient is:

$$\nabla_{\boldsymbol{\theta}}\ell = 2\,(\nabla_{\boldsymbol{\theta}}\boldsymbol{u}_{\boldsymbol{\theta}})^\top\big(V_{\boldsymbol{\theta}}^{\mathrm{EMA}} - \boldsymbol{v}_{\text{cond}}\big)\,. \tag{27}$$

Writing $\boldsymbol{v}_{\text{cond}} = \boldsymbol{v}(\boldsymbol{x}_t, t) + \boldsymbol{v}'$ and noting that $\tilde{\boldsymbol{r}}^{\mathrm{EMA}} := V_{\boldsymbol{\theta}}^{\mathrm{EMA}} - \boldsymbol{v}(\boldsymbol{x}_t, t)$ is deterministic given $\boldsymbol{x}_t$:

$$\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}[\nabla_{\boldsymbol{\theta}}\ell] = 2\,(\nabla_{\boldsymbol{\theta}}\boldsymbol{u}_{\boldsymbol{\theta}})^\top\,\tilde{\boldsymbol{r}}^{\mathrm{EMA}}, \tag{28}$$

since $\mathbb{E}[\boldsymbol{v}' \mid \boldsymbol{x}_t] = \boldsymbol{0}$. The tower property $\nabla_{\boldsymbol{\theta}}\mathcal{L}_{\mathrm{MF}}^{\mathrm{EMA}} = \mathbb{E}_{r,t,\boldsymbol{x}_t}\big[\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}[\nabla_{\boldsymbol{\theta}}\ell]\big]$ gives the stated result. Crucially, no $\boldsymbol{J}\boldsymbol{v}'$ term appears because the EMA tangent is deterministic—compare with the original MeanFlow gradient equation 17 where the stochastic tangent produces the $\boldsymbol{J}\boldsymbol{v}'$ noise. □

C.5 Proof of Proposition 5 (Bias-Variance Tradeoff)

Proposition 6 (Bias-Variance Tradeoff)**.** *The EMA mean-field residual decomposes as*

$$\tilde{\boldsymbol{r}}^{EMA} \;=\; \boldsymbol{r}_{\boldsymbol{\theta}} \;+\; (t{-}r)\,\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t, t, t) - \boldsymbol{v}(\boldsymbol{x}_t, t))\,,$$

where $\boldsymbol{r}_{\boldsymbol{\theta}}$ is the true MeanFlow residual from equation 6. At convergence $\boldsymbol{r}_{\boldsymbol{\theta}} \to \boldsymbol{0}$ and the FM anchor drives $\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t, t, t) \to \boldsymbol{v}(\boldsymbol{x}_t, t)$, giving $\tilde{\boldsymbol{r}}^{EMA} \to \boldsymbol{0}$.

Proof. Expanding $V_{\boldsymbol{\theta}}^{\mathrm{EMA}}$ from equation 8, the value of the compound prediction (ignoring the stop-gradient, which does not affect values) is:

$$V_{\boldsymbol{\theta}}^{\mathrm{EMA}} = \boldsymbol{u}_{\boldsymbol{\theta}} + (t-r)(\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}} \cdot \boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t,t,t) + \partial_t \boldsymbol{u}_{\boldsymbol{\theta}}). \tag{29}$$

The true MeanFlow residual (with the marginal velocity as tangent) is:

$$\boldsymbol{r}_{\boldsymbol{\theta}} = \boldsymbol{u}_{\boldsymbol{\theta}} + (t-r)(\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}} \cdot \boldsymbol{v}(\boldsymbol{x}_t,t) + \partial_t \boldsymbol{u}_{\boldsymbol{\theta}}) - \boldsymbol{v}(\boldsymbol{x}_t,t). \tag{30}$$

Subtracting:

$$\begin{aligned} \tilde{\boldsymbol{r}}^{\mathrm{EMA}} &= V_{\boldsymbol{\theta}}^{\mathrm{EMA}} - \boldsymbol{v}(\boldsymbol{x}_t,t) \\ &= \boldsymbol{r}_{\boldsymbol{\theta}} + (t-r)\,\partial_{\boldsymbol{z}}\boldsymbol{u}_{\boldsymbol{\theta}}(\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t,t,t) - \boldsymbol{v}(\boldsymbol{x}_t,t)). \end{aligned} \tag{31}$$

At convergence, the true residual $\boldsymbol{r}_{\boldsymbol{\theta}} \to \boldsymbol{0}$ (the MeanFlow identity is satisfied) and the FM anchor loss equation 9 supervises the boundary condition, driving $\boldsymbol{u}_{\bar{\boldsymbol{\theta}}}(\boldsymbol{x}_t,t,t) \to \boldsymbol{v}(\boldsymbol{x}_t,t)$ and hence $\tilde{\boldsymbol{r}}^{\mathrm{EMA}} \to \boldsymbol{0}$. ☐

C.6 Derivation of the Trace Weight

Before proving Proposition 7, we derive the trace-weight formula equation 11 as a regularized BLUE-scalar approximation under the isotropic-noise model.

Setup. Under the EMA tangent (Section 3.2.1), proposition 4 and the Reynolds decomposition imply that the per-sample loss conditioned on $\boldsymbol{x}_t$ admits the noise-decomposition

$$\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}\Big[\big\|V_{\boldsymbol{\theta}}^{\mathrm{EMA}} - \boldsymbol{v}_{\mathrm{cond}}\big\|_2^2\Big] \;=\; \underbrace{\big\|\tilde{\boldsymbol{r}}^{\mathrm{EMA}}\big\|_2^2}_{\text{deterministic signal}} \;+\; \underbrace{\mathrm{Tr}\big(\boldsymbol{J}\,\Sigma_{\boldsymbol{v}'}\,\boldsymbol{J}^\top\big)}_{\text{irreducible noise floor } \nu(\boldsymbol{x}_t,r,t)}. \tag{32}$$

The noise floor ν is heteroscedastic: it varies with $(\boldsymbol{x}_t, r, t)$ through both $\boldsymbol{J}$ and $\Sigma_{\boldsymbol{v}'}$, and dominates the per-sample loss in high-curvature regions even though it does *not* contribute to the parameter gradient (its $\boldsymbol{\theta}$-dependence is wiped by the stop-gradient on the JVP).

Heteroscedastic-noise BLUE. Suppose we form a weighted empirical-risk estimator $\hat{\mathcal{L}}(\boldsymbol{\theta}) = \frac{1}{N}\sum_i w_i\,\ell_i(\boldsymbol{\theta})$ from N independent per-sample losses ℓ_i whose noise floors $\nu_i = \mathrm{Var}[\ell_i \mid \boldsymbol{\theta}]$ are heteroscedastic and known. The Best Linear Unbiased Estimator (BLUE) of the population mean assigns

$$w_i^{\mathrm{BLUE}} \;\propto\; \nu_i^{-1}, \tag{33}$$

minimizing the variance of $\hat{\mathcal{L}}$ subject to $\mathbb{E}[\hat{\mathcal{L}}] = \mathbb{E}_{\boldsymbol{x}_t,r,t}\big[\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}[\ell]\big]$. Applying equation 33 to our setting with $\nu_i = \mathrm{Tr}(\boldsymbol{J}_i\Sigma_{\boldsymbol{v}',i}\boldsymbol{J}_i^\top)$ would prescribe $w \propto 1/\mathrm{Tr}(\boldsymbol{J}\Sigma_{\boldsymbol{v}'}\boldsymbol{J}^\top)$. Two practical obstacles remain:

1. The matrix BLUE $w \propto (\boldsymbol{J}\Sigma_{\boldsymbol{v}'}\boldsymbol{J}^\top)^{-1}$ at the gradient level (rather than the trace) requires a per-sample $d \times d$ inversion, costing $\mathcal{O}(d^3)$ per sample.
2. The scalar form $w \propto 1/\mathrm{Tr}(\boldsymbol{J}\Sigma_{\boldsymbol{v}'}\boldsymbol{J}^\top)$ is unbounded as $\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top) \to 0$ (low-curvature regions), where the BLUE prescribes infinite weight on samples with vanishing noise floor.

Isotropic approximation and Tikhonov stabilization. We resolve both pathologies in two steps. First, the isotropic-noise approximation $\Sigma_{\boldsymbol{v}'} \approx \sigma_t^2 \boldsymbol{I}_d$ collapses the trace to a scalar:

$$\mathrm{Tr}(\boldsymbol{J}\Sigma_{\boldsymbol{v}'}\boldsymbol{J}^\top) \;=\; \sigma_t^2\;\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top), \tag{34}$$

which is a single Hutchinson estimate at $\mathcal{O}(d)$ cost. (Appendix D.2 discusses concrete choices of σ_t.) Second, we add a Tikhonov regularizer of magnitude d to the denominator:

$$w(\boldsymbol{x}_t, r, t) \;:=\; \frac{d}{d + \sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)} \;=\; \frac{1}{1 + \sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)/d}, \tag{35}$$

matching equation 11. The regularization has three desirable properties:

- **Boundedness.** $w \in (0,1]$ for all $(\boldsymbol{x}_t, r, t)$, with $w = 1$ exactly when $\sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top) = 0$ (no reweighting in flat regions).

- **BLUE limit.** As $\sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)\to\infty$, $w\to d/[\sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)]$, recovering the BLUE-scalar prescription equation 33 up to a constant.
- **Bias-variance tradeoff.** The added d trades a small estimator bias (the BLUE limit is approached but not attained) for bounded weights, eliminating the divergence at low curvature.

The choice of regularizer magnitude d has a clean physical reading: under the BLUE limit the noise contribution per sample equals $w\cdot\sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)=d$ (i.e., one unit per data dimension), and the regularizer is the smallest constant for which $w\leq 1$ uniformly.

C.7 Proof of Proposition 7 (Trace-Weight Properties)

Proposition 7 (Trace-Weight Properties)**.** *Under the trace-weight loss equation 10 and the isotropic approximation $\Sigma_{\boldsymbol{v}'}\approx\sigma_t^2\boldsymbol{I}_d$: **(a)** the mean-parameter gradient is the trace-weighted form of proposition 4 with w as a stop-gradient scalar, so the BLUE-style normalization does not enter the parameter update; **(b)** the expected per-sample contribution to the loss decomposes into a deterministic signal $w\,\|\tilde{\boldsymbol{r}}^{EMA}\|_2^2$ and a constant noise floor $w\,\sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)=d\,\sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)/(d+\sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top))\leq d$, eliminating the unbounded curvature growth from theorem 2; **(c)** w is monotonically decreasing in $\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)$ and, by theorem 6, in $(t-r)^2\|\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}\|^2$, so high-curvature long-interval samples are automatically downweighted—subsuming inverse-variance weighting, progressive scheduling, and acceleration-aware interval sampling in a single per-sample scalar.*

Proof. **(a) Gradient form.** The trace weight w in equation 11 is wrapped in a stop-gradient operator on $\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)$, so $\nabla_{\boldsymbol{\theta}}w=0$ and w acts as a fixed scalar multiplier. Differentiating the trace-weighted MSE in equation 10 therefore gives

$$\nabla_{\boldsymbol{\theta}}\mathcal{L}_{\mathrm{VaMF}}^{\mathrm{MF}} \;=\; 2\,\mathbb{E}\big[w\,(\nabla_{\boldsymbol{\theta}}\boldsymbol{u}_{\boldsymbol{\theta}})^\top\,\tilde{\boldsymbol{r}}^{\mathrm{EMA}}\big], \tag{36}$$

identical to proposition 4 except that each per-sample contribution carries the scalar weight w. The conditional fluctuation $\boldsymbol{v}'$ produces a cross term $\mathbb{E}[(\nabla_{\boldsymbol{\theta}}\boldsymbol{u}_{\boldsymbol{\theta}})^\top\boldsymbol{v}']=\boldsymbol{0}$ that vanishes by $\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}[\boldsymbol{v}']=\boldsymbol{0}$, so the BLUE-style normalization does not propagate into the gradient computation.

(b) Bounded noise floor. Writing $V_{\boldsymbol{\theta}}^{\mathrm{EMA}}-\boldsymbol{v}_{\mathrm{cond}}=\tilde{\boldsymbol{r}}^{\mathrm{EMA}}-\boldsymbol{v}'$ with the EMA tangent making $\tilde{\boldsymbol{r}}^{\mathrm{EMA}}$ deterministic given $\boldsymbol{x}_t$, the inner expectation of the per-sample loss is

$$\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}\big[w\,\|V_{\boldsymbol{\theta}}^{\mathrm{EMA}}-\boldsymbol{v}_{\mathrm{cond}}\|_2^2\big] \;=\; w\,\|\tilde{\boldsymbol{r}}^{\mathrm{EMA}}\|_2^2+w\cdot\sigma_t^2\,\mathrm{Tr}(\Sigma_{\boldsymbol{v}'}), \tag{37}$$

where the cross term vanishes as in (a). Substituting the isotropic-noise approximation $\Sigma_{\boldsymbol{v}'}\approx\sigma_t^2\boldsymbol{I}_d$ and the explicit form of w yields the noise contribution

$$w\cdot\sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top) \;=\; \frac{\sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)}{1+\sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)/d} \;=\; \frac{d\,\sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)}{d+\sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)} \;\leq\; d, \tag{38}$$

with equality only in the limit $\sigma_t^2\,\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)\to\infty$. The trace weight thus caps the noise term at the data dimensionality, eliminating the unbounded curvature growth from theorem 2.

(c) Curvature-aware downweighting. The map $w(\xi)=1/(1+\sigma_t^2\xi/d)$ is strictly decreasing in $\xi=\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)\geq 0$. By theorem 6 and the Reynolds decomposition, $\mathbb{E}[\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)]$ scales with $(t-r)^2\|\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}\|^2$ to leading order, so w shrinks on the same factor and high-curvature long-interval samples are automatically downweighted—realizing the curvature-aware schedule prescribed in Section 3.2.3 in a single per-sample scalar without requiring an explicit progressive schedule or interval-importance sampling. $\square$

D Additional Results

D.1 Acceleration-to-Noise Ratio from the Trace Weight

The trace weight in proposition 7 reweights samples by an empirical proxy for the squared acceleration-to-noise ratio of the curvature-variance principle. The connection becomes precise when the schedule σ_t is chosen consistently with the actual conditional-velocity covariance.

Proposition 8 (ANR from the Trace Weight)**.** *Let* $\rho(\boldsymbol{x}_t, r, t) := \sigma_t^2 \operatorname{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)/d$ *denote the curvature term in the denominator of equation 11, and assume the schedule satisfies* $\sigma_t^2 = \operatorname{Tr}(\Sigma_{\boldsymbol{v}'})/d$ *(the moment-matched isotropic approximation, achieved exactly by the* BLUE-GAUSS *schedule of Appendix D.2 under Laplace's approximation of* p_{data}*). Then near convergence,*

$$\rho \approx \frac{\|\Delta(\boldsymbol{x}_t, r, t)\|^2}{\operatorname{Tr}(\Sigma_{\boldsymbol{v}'})} \approx \frac{(t-r)^2}{4} \cdot \frac{\left\|\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}(\boldsymbol{x}_t, t)\right\|^2}{\operatorname{Tr}(\Sigma_{\boldsymbol{v}'})} = \frac{\operatorname{ANR}(\boldsymbol{x}_t, r, t)^2}{4}, \tag{39}$$

where $\operatorname{ANR} := (t-r)\left\|\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}\right\| / \sqrt{\operatorname{Tr}(\Sigma_{\boldsymbol{v}'})}$*. The trace weight is then* $w = 1/(1+\rho) = 1/(1+\operatorname{ANR}^2/4)$.

Proof. **Step 1: noise floor and curvature gap.** Under the EMA tangent (Section 3.2.1) and at convergence, proposition 5 gives $\tilde{\boldsymbol{r}}^{\text{EMA}} \approx \mathbf{0}$, so the per-sample loss reduces to its noise floor $\operatorname{Tr}(\boldsymbol{J}\Sigma_{\boldsymbol{v}'}\boldsymbol{J}^\top)$ (cf. equation 32). Substituting the Reynolds-residual expansion $\boldsymbol{r}_{\boldsymbol{\theta}}^{\text{sg}} = \Delta + (t-r)\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{u}_{\boldsymbol{\theta}}$ from Section 3.1 and using the MeanFlow identity $(t-r)\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{u} = -\Delta$ gives the per-sample noise floor $\left\|\boldsymbol{J}\boldsymbol{v}'\right\|_2^2$ at first order; taking expectations and applying the trace identity yields

$$\mathbb{E}_{\boldsymbol{x}_0|\boldsymbol{x}_t}\big[\|\boldsymbol{J}\boldsymbol{v}'\|_2^2\big] = \operatorname{Tr}\big(\boldsymbol{J}\,\Sigma_{\boldsymbol{v}'}\,\boldsymbol{J}^\top\big) \approx \left\|\Delta(\boldsymbol{x}_t, r, t)\right\|^2 \tag{40}$$

in the small-gap regime.

Step 2: isotropic moment matching. Under the schedule assumption $\sigma_t^2 = \operatorname{Tr}(\Sigma_{\boldsymbol{v}'})/d$, the isotropic-noise approximation equation 34 gives $\operatorname{Tr}(\boldsymbol{J}\Sigma_{\boldsymbol{v}'}\boldsymbol{J}^\top) = \sigma_t^2 \operatorname{Tr}(\boldsymbol{J}\boldsymbol{J}^\top) = \operatorname{Tr}(\Sigma_{\boldsymbol{v}'})\operatorname{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)/d$. Combining with equation 40,

$$\frac{\sigma_t^2 \operatorname{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)}{d} \approx \frac{\|\Delta\|^2}{\operatorname{Tr}(\Sigma_{\boldsymbol{v}'})} \iff \rho \approx \frac{\|\Delta\|^2}{\operatorname{Tr}(\Sigma_{\boldsymbol{v}'})}. \tag{41}$$

Step 3: substitute the flow-acceleration expansion. Theorem 6 gives $\|\Delta\| \approx \frac{t-r}{2}\left\|\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}\right\|$ to leading order, so $\|\Delta\|^2/\operatorname{Tr}(\Sigma_{\boldsymbol{v}'}) \approx (t-r)^2\left\|\frac{\mathrm{D}\boldsymbol{v}}{\mathrm{D}t}\right\|^2/[4\operatorname{Tr}(\Sigma_{\boldsymbol{v}'})] = \operatorname{ANR}^2/4$, completing equation 39. □

Schedule sensitivity. The proof requires the moment-matched schedule $\sigma_t^2 = \operatorname{Tr}(\Sigma_{\boldsymbol{v}'})/d$. Practical schedules (Appendix D.2) only approximate this:

- BLUE-GAUSS matches it exactly under Laplace's approximation of p_{data}, so equation 39 holds to leading order.
- U-SHAPE qualitatively tracks the location of the peak of $\operatorname{Tr}(\Sigma_{\boldsymbol{v}'})$ (eq. (43)) but not its absolute scale, so the connection holds up to a t-dependent multiplicative constant.
- LINEAR and QUADRATIC are monotone in t and *do not* approximate the U-shape; the connection in equation 39 is then valid only as a qualitative trend rather than a quantitative identity.

The trace weight remains a valid BLUE-scalar approximation in all three cases (Appendix C.6); only the ANR *interpretation* of ρ is schedule-sensitive.

D.2 Schedule σ_t in the Trace Weight

The trace weight equation 11 introduces a single hyperparameter, the time-dependent scalar σ_t that approximates the conditional-velocity standard-deviation scale via $\Sigma_{\boldsymbol{v}'} \approx \sigma_t^2 \boldsymbol{I}_d$. The exact $\Sigma_{\boldsymbol{v}'}$ depends on p_{data} and is not generally closed-form; we first derive a tractable instance under Laplace's approximation $p_{\text{data}} \approx \mathcal{N}(\mathbf{0}, \sigma_d^2 \boldsymbol{I}_d)$, then compare five practical schedules empirically.

Conditional-velocity variance under Laplace's approximation. Following Laplace's approximation in Bayesian statistics [33, 34], we replace p_{data} by a Gaussian centered at the mode with covariance set to the inverse Hessian of $-\log p_{\text{data}}$ at that mode. Take $\boldsymbol{x}_0 \sim \mathcal{N}(\mathbf{0}, \sigma_d^2 \boldsymbol{I}_d)$ and $\boldsymbol{x}_1 \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I}_d)$ with the linear interpolant $\boldsymbol{x}_t = (1-t)\boldsymbol{x}_0 + t\boldsymbol{x}_1$. Conditional on $\boldsymbol{x}_0$, the noise contribution is $\boldsymbol{x}_t \mid \boldsymbol{x}_0 \sim \mathcal{N}((1-t)\boldsymbol{x}_0, t^2\boldsymbol{I}_d)$. Combining this Gaussian likelihood with the Gaussian prior on $\boldsymbol{x}_0$ via conjugate inference,

$$\operatorname{Var}[\boldsymbol{x}_0 \mid \boldsymbol{x}_t] = \frac{t^2\sigma_d^2}{\sigma_d^2(1-t)^2 + t^2}\,\boldsymbol{I}_d. \tag{42}$$

Substituting $\boldsymbol{v}_{\text{cond}} = \boldsymbol{x}_1 - \boldsymbol{x}_0 = (\boldsymbol{x}_t - \boldsymbol{x}_0)/t$ gives

$$\Sigma_{\boldsymbol{v}'}(\boldsymbol{x}_t, t) \;=\; \frac{1}{t^2}\,\mathrm{Var}[\boldsymbol{x}_0 \mid \boldsymbol{x}_t] \;=\; \frac{\sigma_d^2}{\sigma_d^2(1-t)^2 + t^2}\,\boldsymbol{I}_d. \tag{43}$$

Equation (43) is U-shaped in t: at $t = 0$ and $t = 1$ it equals σ_d^2 (resp. 1 for the unit-variance case $\sigma_d = 1$), and at $t = 0.5$ it reaches its maximum of $2\sigma_d^2/(\sigma_d^2 + 1)$. The peak coincides exactly with the regime where the Jacobian-deviation norm $\mathrm{Tr}(\boldsymbol{J}\boldsymbol{J}^\top)$ from theorem 2 is largest, since both reflect the same physical phenomenon—mid-t mode mixing.

Practical schedules. Real p_{data} (e.g. image latents) is non-Gaussian and Eq. equation 43 is at best a first-order approximation. We therefore benchmark five practical schedules:

NONE. $\sigma_t = 1$. The bare BLUE-scalar form. Serves as the cleanest test of the curvature term in isolation.

LINEAR. $\sigma_t = t$. The simplest monotone schedule with $\sigma_t^2 = t^2$ in the denominator of equation 11. Suppresses high-t samples where the noise has dispersed.

QUADRATIC. $\sigma_t = t^2$, so $\sigma_t^2 = t^4$. Steeper monotone profile, used in early experiments.

U-SHAPE. $\sigma_t = \sqrt{t(1-t)}$, so $\sigma_t^2 = t(1-t)$. Peaks at $t = 0.5$ and vanishes at the endpoints—qualitatively matching the location of equation 43's peak without committing to a parametric form.

BLUE-GAUSS. $\sigma_t^2 = 1/(\sigma_d^2(1-t)^2 + t^2)$ from equation 43 with $\sigma_d = 1$. The closed-form BLUE under Laplace's approximation of p_{data}.

The toy benchmark sweep across all five schedules appears in Section 5, Figure 5; the empirical winner ($\sigma_t = 1$) is used as the default in the main text and in Appendix E.

E DiT Latent-Space Training on ImageNet

We test whether the curvature-variance principle scales beyond toys by training DiT [44] variants on ImageNet in the latent space of a frozen Stable Diffusion VAE, following the recipe of [24]. Both baselines and VaMF runs share identical hyperparameters and code paths; only the loss differs (vanilla MeanFlow MSE versus the trace-weighted MSE in equation 10 with $\sigma_t = 1$, the toy-sweep winner of Figure 5, and a single Hutchinson probe).

E.1 DiT-B/4, ImageNet-256×256

Setup: logit-normal (r, t) sampling with mean -0.4 and stddev 1, overlap rate 0.75, AdamW with constant learning rate 10^{-4}, EMA decay 0.9999, batch size 256 across 32 TPU v4 chips, 300,000 steps. The single-probe Hutchinson estimator adds approximately one additional JVP per step, which we measure as a $\sim 22\%$ wall-clock overhead on TPU v4 (2.20 steps/s for VaMF-TW versus 2.65 steps/s for the baseline). At identical optimizer step counts, VaMF-TW maintains a small but consistently lower per-sample loss than the baseline beyond $\sim 20{,}000$ steps. The matched-step loss comparison in Table 6 (baseline run `fidpdet7` and VaMF-TW run `ymxp1xfg` on Weights&Biases) reflects the latest snapshot before submission; the VaMF-TW run is on track to complete the full 300,000 steps shortly after the abstract deadline, at which point we will add the $\text{FID}_{50\text{k}}$ comparison to the camera-ready version.

Table 6: Matched-step training-loss comparison on the DiT-B/4 ImageNet-256 latent run. Each cell averages the per-step training MSE over the 30 logged points immediately preceding the listed step. Negative deltas indicate VaMF-TW is lower; the trace weight provides a small but consistent improvement once it becomes load-bearing past step ~ 20k.

Step	MeanFlow (`fidpdet7`)	VaMF-TW (`ymxp1xfg`)	Δ
10,000	3,718	3,736	$+0.5\%$
20,000	3,827	3,759	-1.8%
30,000	3,888	3,884	-0.1%
35,500	3,953	3,885	-1.7%